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Multimodal Teleoperation of a Wearable Dual-Arm Supernumerary Robotic System

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Abstract

The base of a supernumerary robotic limb is a person. This project built a teleoperation system for two seven-degree-of-freedom arms carried on a worn backpack frame and commanded by a *second* person, gave it a safety layer appropriate to a manipulator standing beside somebody who is not commanding it, and ran a first pilot comparing direct control against shared-autonomy assistance.

A body-worn master supplies position from whichever sensor observes each component most directly, and classifies all fourteen channels so a failing sensor degrades the system announcedly rather than silently. Six of fourteen met the health criterion; failing ones injected 54.5 mm of unrequested motion. Vertical and fore-and-aft command track correctly; lateral does not, and no axis mapping fixes it. The two arms share no reachable point under the as-built mount; a costed mount change raised home clearance from 0.1610 m to 0.2202 m. The task set solves 13 927 inverse-kinematics calls with no failures; replacing the motion generator removed 51.3 mm of unrequested excursion; and perception on four physical cameras fits a work surface to 1.43 mm RMS.

Under ethical approval (SETREC 7111986), five operators completed twenty-three sessions comparing direct VR control against shared-autonomy assistance: assisted sessions averaged shorter completion times and fewer re-grips but more hand travel, with sub-5 mm lag throughout. The pilot has no separate wearer and is pilot-scale; nothing was validated on a loaded physical arm, and the two-person study this system was built for remains the target for the work that follows.

Acknowledgements

I thank my supervisor, Dr. Dandan Zhang, for the direction of this project and for consistently pushing the work toward what could be measured rather than what could be claimed.

I thank my co-supervisor, Dr. Etienne Burdet, for the framing that changed the shape of the project: treating the wearer and the operator as two people with different stakes, rather than as one role, is what turned a control problem into a question worth asking.

I am grateful to the Human Robotics Group for access to the Dr. Octopus platform, and to everyone who kept two Kinova arms and a backpack frame available and working. Finally, I thank my family and friends for their support over the course of the year.

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Chapter 1

Introduction

A supernumerary robotic limb is an extra arm worn on the body, working alongside the wearer’s own limbs rather than replacing or rehabilitating them [YHC⁺21, P⁺21]. What separates such a device from a fixed manipulator is consequential for control: the base of the robot is a person. It accelerates, sways and steps, and it cannot be told to hold still.

This thesis reports work on the MUlti-limb Virtual Environment, known within Imperial College London as the Dr. Octopus platform [Imp], which mounts two Kinova Gen3 seven-degree-of-freedom arms [Kin22] on a backpack frame and commands them from a *second* person: an operator who cannot feel the arms, controlling them for a wearer who cannot command them. No prior work has measured both what shared autonomy buys such an operator and what it costs the person carrying the arms, in one system, under the same trial (Chapter 2); that gap is what this project is built to close.

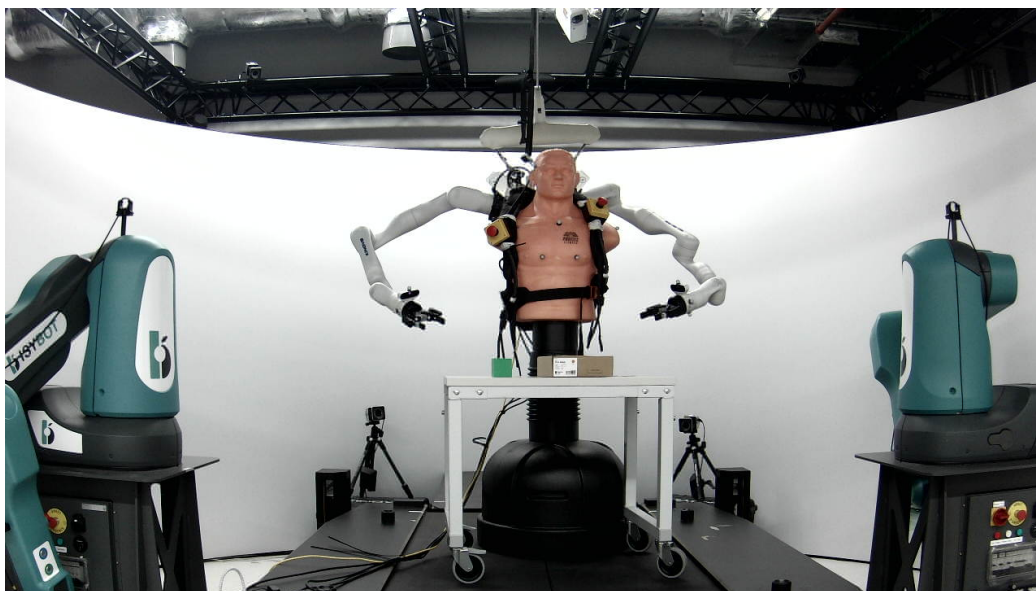


Figure 1.1: The platform as built: two Kinova Gen3 arms on a backpack frame, on a torso mannequin, photographed by the scene camera. The two teal columns are the laboratory’s fixed manipulators and are not part of this system.

Two premises shaped the engineering. First, commanding fourteen joints through a body-worn master with no force feedback is a usability problem before a control one, so position comes from whichever sensor observes each component best, not the slave’s full kinematic chain. Second, a body-worn master *will* lose channels over its life, so the system treats channel loss as an operating condition: every channel carries a health verdict, and when

too many are lost it reports position unavailable rather than fabricating one – on a manipulator beside a person’s head, worse than none.

1.1 Objectives

1. Design and build an instrumented mannequin master reproducing the Gen3 morphology at reduced scale, characterised against recorded data.
2. Derive a master-to-slave command mapping that degrades predictably as sensing channels are lost, and quantify what each loss costs.
3. Implement a safety architecture appropriate to a manipulator mounted beside a person, validated by fault injection rather than inspection.
4. Characterise the reachable workspace of the mounted configuration and determine which manipulation tasks it can support.
5. Establish a verification methodology producing traceable evidence for every claim, and apply it throughout.
6. Run a pilot comparison of direct control against shared-autonomy assistance with real operators, and identify what stands between it and the full two-person study the architecture was designed for.

The planning report framed this project around a haptic master with physical springs and virtual slave-side impedance. That objective was not delivered: [Chapter 5](#) explains why, and [Chapter B](#) records the route in full.

1.2 Contributions

Each is stated with its evidence class (implemented, verified in simulation, measured on hardware, or measured on a pilot session) in [Chapter 4](#): a graceful-degradation architecture for a body-worn master; a reduced-degree-of-freedom command mapping; a safety architecture in which every blocking mechanism is observable; a workspace characterisation with a quantified, buildable mount correction; a jerk-limited motion generator; a calibrate-then-plan perception stage exercised on four physical cameras; and a first pilot of the direct-versus-shared-autonomy comparison with five operators, run under ethical approval once VR controllers supplied a working alternative to the master’s incoherent channels. **No contribution below the pilot was validated on physical hardware carrying a load.**

The single most transferable outcome does not appear in that list. Twenty-six times across this project, a result that looked like a system failure was a defect in the measuring tool, and validating an instrument against a known answer before trusting a surprising reading it produces is the practice applied to every number in this report ([Section 5.3](#)).

1.3 Report structure

[Chapter 3](#) summarises what was designed, built and run; [Chapters A](#) and [C](#) to [E](#) carry the full engineering detail. [Chapter 4](#) reports the headline measurements and the pilot; [Chapters F](#), [H](#) and [I](#) carry the full campaign. [Chapters 5](#) and [6](#) interpret the results and close against the objectives above; [Chapter J](#) is the un-run two-person protocol this project was built for, and [Chapter L](#) the pilot session in full.

Chapter 2

Background

Five bodies of work bear on the system built here.

Supernumerary robotic limbs. Parietti and Asada’s founding demonstrations brace a limb against the environment to stabilise the wearer’s posture [PA14, PA13], which turns the wearer’s body from a disturbance source into part of a closed chain. Later reviews map the design space along consistent axes – mounting location trades reach against the moment arm imposed on the wearer, actuation trades bandwidth against worn mass – and note that the field reports embodiment far more often than task performance [YHC⁺21, P⁺21, Z⁺26]. Worn mass binds hardest: each Gen3 masses 8.2 kg, so the pair with harness exceeds 17 kg, and simulated wearer loads suggest the biomechanical cost is not negligible [MK24].

The operator and the wearer as different people. Fusion mounts a robotic head and two arms on a worn frame, driven by a remote operator [SSM⁺18] – but its three modes are levels of *physical coupling*, not machine autonomy, and the publication reports no user study. Work on near-body interaction found that raising *apparent* autonomy (a concealed operator, a wearer with no task of their own) raised physiological arousal and lowered self-reported trust [ZGD⁺26]: the wearer’s experience is not a free variable, though this is not evidence about real autonomy or an occupied wearer.

Master interfaces for dissimilar kinematics. Guggenheim and Asada argue for miniature physical masters on the grounds that geometric similarity supplies feedback inherently [GA21]; the argument covers similarity of *form*, not *scale*, and this system’s master-to-slave reach ratio is roughly one to three. Joint-space correspondence preserves posture but needs every master joint measured accurately; task-space correspondence decouples the chains at the cost of the slave’s redundant degree of freedom.

Shared autonomy. Dragan and Srinivasa formalise arbitration as policy blending, weighted by prediction confidence [DS13]; Javdani et al. recast it as a POMDP over the user’s goal and show that assistance which does not wait for high confidence helps operators finish faster [JSB15]. Neither was developed for a robot mounted on a second person, where the cost of assisting toward the wrong goal is not a slower trial but a manipulator moving toward someone who did not command it.

Measuring embodiment, workload and trust. Ownership over an artificial limb is measurable and decomposes into ownership, agency and location, which can move independently [BC98, LSK⁺08]. Workload separates into the NASA Task Load Index’s six subscales [HS88]; physical exertion is better tracked with a Borg scale [Bor82], appropriate for an uncompensated master arm or a carried mass.

2.1 The gap

No study measures the operator and the wearer together, under the same autonomy, in the same trial: shared autonomy has been shown to help an operator where the robot is attached to nobody, and raising apparent autonomy has been shown to lower a wearer's trust where the autonomy was a concealed human. Until both are measured in one system, the exchange rate between what shared autonomy buys the operator and what it costs the person carrying the arms is unknown. [Chapter 1](#) states the aim this gap sets; the full review, with every citation's scope stated precisely, is in [Chapter M](#).

Chapter 3

Methods

This chapter summarises what was designed, built and run. Full engineering detail is carried in [Chapters A and C](#) to [E](#) so it can be checked and reproduced without lengthening the argument here.

3.1 The master arm

The master is a miniature mannequin reproducing the Gen3’s joint sequence at roughly a third of its scale ([Fig. A.1, Chapter A](#)), moved with the fingers of one or both hands. A Teensy 4.1 reads fourteen rotary potentiometers, two inertial units, two force-sensitive resistors and two buttons, emitting one text frame at 100 Hz ([Table A.1](#)). Potentiometer counts pass through an integer exponential moving average ([Eq. \(A.1\)](#)) before conversion to degrees; the inertial sensors are read as a single burst so all six axes come from one sampling instant.

Each channel carries a health verdict computed from its own recent history: *dead*, *incoherent* (present but not tracing a trajectory), *intermittent*, or *alive*. Validation is scoped to the channels the active mode actually consumes, so a dead channel the mode does not read cannot reject a frame, and when no well-conditioned channel remains the system reports position *unavailable* rather than extrapolating one.

3.2 From master pose to arm command

The commanded position is built in spherical form about the master’s shoulder rather than by forward kinematics from seven joint angles ([Eqs. \(C.1\) to \(C.3\)](#)): elevation from the gravity vector, azimuth from the shoulder roll, and radius from the forward-kinematic magnitude of the bend joints. Each component is taken from whichever sensor observes it most directly, so a dead roll channel can be passed through as zero without corrupting the radius. The commanded end-effector position is an anchor plus a scaled displacement from a reference latched at clutch engagement ([Eq. \(C.4\)](#)), so releasing and re-engaging the clutch indexes like a mouse lifted and set down again, and the reachable set is bounded by the robot rather than by the master’s own range.

Table 3.1: Operating modes, condensed from [Table C.1](#). Autonomy runs at a quarter of teleoperation’s speed by design: nobody is holding a master, so the only bound on a wrong motion is how long it takes to happen.

Mode	Input	Robot decides	$v_{\max}$
01/02 direct	mannequin or VR controllers	nothing	0.60 rad/s
03/04 shared autonomy	master or controllers, vision	wrist orientation	0.60 rad/s
06 full autonomy	typed or spoken goal	target, approach, grasp, place	0.15 rad/s

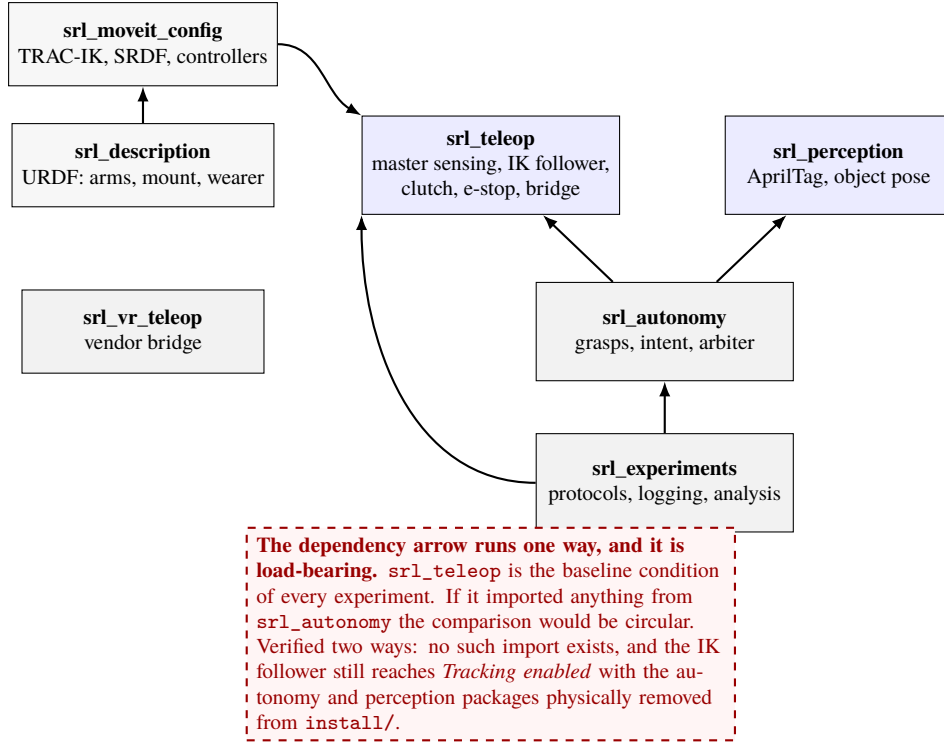


Figure 3.1: The package decomposition and its one-way dependency graph: teleoperation, the baseline condition of every planned experiment, imports nothing from autonomy, verified by source inspection and by the follower reaching its running state with autonomy and perception deleted from the install tree.

Five modes are delivered (Table 3.1); one safety stack serves all of them, and dropping any shared mechanism is refused regardless of mode. The laboratory’s actual arrangement seats the operator across the room from the wearer, holding the VR controllers as a six-degree-of-freedom input device and watching the robot directly rather than wearing the headset (Section C.2.1); the operator-to-robot alignment is a measured yaw, never a mirror, since facing someone and copying them is a reflection that would invert every commanded orientation while positions looked correct.

3.2.1 Safety architecture

Collision avoidance is graduated in three stages (Section C.1.1): collision-aware inverse kinematics, redundancy re-seeding on rejection, and a geometric clearance floor of 150 mm measured from the transform tree – not the semantic collision description, which excludes the proximal pairs a shoulder mount actually threatens. Every mechanism able to stop the arm registers by name, must declare how it clears, and expires into an explicit *unknown* state (still blocking) if its own loop stops running, so a blocker cannot silently keep a stale “clear.” Three independent triggers latch the emergency stop until reset: an explicit call, both master buttons held together, and a dead-man keyed on the master data’s *source* timestamp rather than message arrival, since a degraded master keeps publishing stale values at full rate rather than falling silent.

3.3 Perception, planning and motion

A calibrate-then-plan stage (Chapter D) replaced declared object and surface coordinates with measured ones: a wrist-camera sweep holding one attitude per pass, printer-bed-probe style, fits the working surface and seg-

ments objects from the image, and hands the result to an RRT-Connect planner [KL00] evaluated against the safety layer’s own clearance. Detection is open-vocabulary (YOLO-World [CSG⁺24] with FastSAM [ZDA⁺23]), falling back to a hue-and-depth filter; grasp synthesis is the geometric principal-axis method of Eqs. (D.1) and (D.2) rather than a learned one, chosen because it can *name* a refusal’s cause and fits the 4 GiB of video memory on this machine.

The per-joint step limiter that previously was the entire motion generator left the commanded hand 51.3 mm off the path its own inverse kinematics implied (Section E.1); replaced with Ruckig [BK21], a jerk-limited generator driving every joint from a single scalar phase (Eq. (E.1)), so the commanded configuration is on the line between endpoints by construction. Replaying recorded joint commands onto the physical arm shows every joint parks 0.305° short of its target on the side it approached from (Section E.3); the two candidate models – a terminal offset and a proportional gain error – are distinguishable only at short travel, and propagating the error through the manipulator Jacobian selects the terminal offset at $r = 0.923$.

3.4 Verification methodology

Twenty-six times in this project a result that looked like a system failure was a defect in the measuring tool (Chapter G, Section 5.3). The practice applied throughout: every analysis script carries a known-answer test, synthetic data only where its ground truth is *constructed* rather than rendered, and a result contradicting an earlier measurement is an instrument check until reconciled. Because the solver restarts randomly, every reachability claim here is ten repeats over a path densified to 20 mm, not single endpoint calls, and a pose not solid on all ten is not reported as reachable.

3.5 Ethics and the pilot study

This study was approved by Imperial College London’s Science, Engineering and Technology Research Ethics Committee, approval number 7111986. All participants gave informed consent before taking part. Participant data are anonymised throughout as P1–P5 and M1–M3; `write_manifest()` refuses to record a name, email, date of birth, address or telephone number, so anonymity is enforced in code, not only in this text.

The full two-person protocol (Chapter J) crosses direct teleoperation against shared autonomy over five tasks with a separate wearer, designed in full before this report but not run: it depends on master channels that were incoherent (Chapter H) and an approved protocol that did not yet exist. Both have since partly cleared – approval was obtained, and VR controllers (Table 3.1) supply a command that does not depend on the master’s damaged channels – making a first, smaller pilot practical ahead of a full session with a separate wearer.

Five operators (P1–P5) each completed between two and seven recorded sessions, performing manipulation-style tasks – object tracking, target reaching, position matching – once under direct VR control and once with shared-autonomy assistance, using the desk arrangement of Section C.2.1. Coverage is uneven, which Section 4.4 reports rather than hides. Every sample was logged automatically at 20 Hz – pose, clutch engagement, VR spatial lag – with an event log of every clutch and emergency-stop transition, giving completion time, hand travel, re-grips, active-control fraction and mean lag with no manual timing. A further three operators (M1–M3) attempted the same protocol from the mannequin master. This pilot measures *operator* performance only; it has no separate wearer and so does not address the wearer-side hypotheses (H3, H4) of the full protocol, which remain for the session Chapter J specifies.

Chapter 4

Results

Every number below carries its strength of evidence, and **no result before the pilot in [Section 4.4](#) was validated on a physical Kinova arm carrying a load**: most of this project is simulated against mock hardware, plus recorded data from the physical master and four cameras ([Chapter H](#): full derivations).

4.1 The command mapping and its safety layer

Six of fourteen master channels are alive; the per-channel verdicts, the 54.5 mm of unrequested motion the incoherent ones inject, and the repair that follows from it are moved to [Section H.2](#) in full, alongside the degradation architecture’s own discussion ([Section N.1](#)) – this section carries only what the safety layer itself did, live, on the pilot below.

The safety layer handled fourteen of fourteen injected faults (channel dropout, emergency stop mid-motion, wrong-object latching, stale detections, clock non-monotonicity and more); the two that needed several attempts both traced to a mechanism keyed on the wrong observable, fixed by reading the master data’s source timestamp rather than its arrival, and by parking the arm before talking to the driver rather than after.

The pilot ([Section 4.4](#)) is the only source of *live* evidence for this layer: across all twenty-six recorded sessions the geometric clearance floor – the mechanism that would hold an arm 150 mm short of the wearer – never activated once, meaning no recorded operator ever drove either arm inside that margin. The manual/dead-man emergency stop did fire for real: four holds across three sessions, one tripping twice, the first held 22.6 s *mid-session* (67 % through) with clutch engagement resuming around it – not an end-of-recording gesture. The geometric path remains validated only against synthetic fault injection.

4.2 Workspace and motion

Over a 63-cell grid spanning the frontal working volume with both arms verified at the home pose, **no cell was reachable by both arms** simultaneously ([Fig. 4.1](#)), and the result survives freeing the wrist yaw and adding solver retries.

Moving the mounts inboard, forward and down with extra forward tilt raises shared cells to 22 of 63 within a buildable 0.20 m separation. That change was applied, raising home-pose clearance from 0.1610 m (a reading of the mount itself, since no joint moves it) to 0.2202 m, and moving pick-and-place targets from 0.530 m off centre to 0.290 m. Neither arm crosses the body centre line at any surface height or distance in 3360 swept configurations; the constraint is the wearer’s torso, not arm posture.

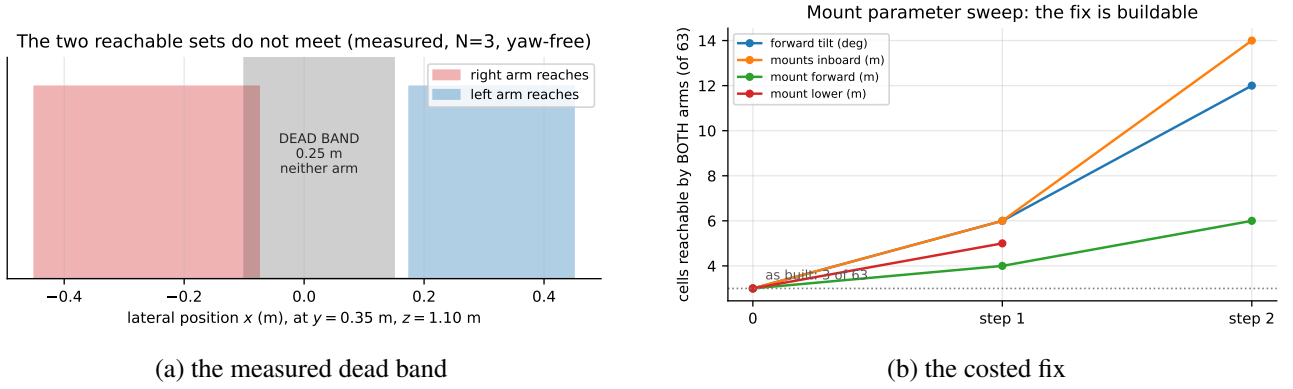


Figure 4.1: (a) The two reachable sets do not meet. (b) A mount-placement sweep: forward tilt and inboard translation help, splaying outward *hurts*. Kinematic only, so these are upper bounds pending a clearance check.

The per-joint step limiter that constituted the whole of teleoperation motion generation, measured against a 30 mm grasp capture gate, is compared with its jerk-limited replacement in Table 4.1: it left the hand off the solved path by nearly twice the gate, at over twelve times the manufacturer’s declared joint speed, because it never read the limits file.

Table 4.1: The same commanded slew, both arms, at the master’s own 50 Hz. Off-path is the largest distance the hand leaves the solver’s straight line.

Generator	Off-path	Peak $\dot{q}$ of limit	Arrival
Per-joint clamp, as shipped	51.3 mm	12.53×	2 cycles
Jerk-limited, delivered	0.0 mm	1.00×	1 cycle

Replaying recorded joint commands on the physical arm shows every joint parks 0.305° short of its target, on the side it approached from – a terminal offset, confirmed by propagating the error through the manipulator Jacobian ($r = 0.923$) rather than a proportional-gain model. A single scalar per arm, applied as overshoot in the direction of travel, removes 74 to 76 % of it. Folded into the positioning error budget with a re-derived pad offset and a measured-surface centring correction, this takes the combined budget from 18.64 to 12.91 mm RSS and 33.39 to 19.99 mm worst case – inside the 30 mm requirement, *provided* the still-unmeasured camera-to-wrist transform is small. The first sustained attempt to drive a physical arm from its own wrist camera measured that transform directly and found it is not: four poses of a stationary table disagreed by 8.45° , 34 to 59 mm of position error over the working range (Section H.7).

4.3 Perception: what the cameras actually detected

Instrumented at thirty-three stages per camera and pointed at four physical cameras for the first time (Chapter F), the wrist cameras fit a support plane at 1.43 mm RMS with 74 % inliers at working distance – within a factor of three of the same code against a renderer, the first evidence the surface-fitting stage transfers to hardware. Segmentation is scoped to the two objects the task actually uses, not the whole scene (Fig. 4.2): the cube is isolated by the project’s own colour classifier, and the box – outside that palette – by thresholding on its own measured hue, since applying the cube’s palette to it misfires on the box’s brown tones. Both recover as oriented boxes rather than axis-aligned ones, which is what makes a turned object’s true width – the quantity the grasp planner needs – available at all.

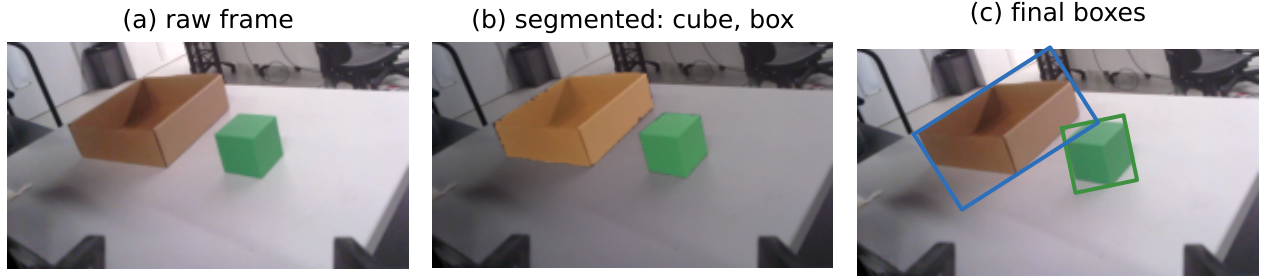


Figure 4.2: The left-hand wrist camera’s raw frame (a), the cube and box segmented from it (b), and the oriented boxes recovered from each mask (c) – the same real frame at every stage, not three different examples.

Depth, not colour, is what a box is drawn from (Fig. 4.3): the same two objects, boxed and range-labelled in their own depth map. Table 4.2 collects this frame’s numbers against the plane-fit method of Fig. 4.2; the two disagree by tens of millimetres because they measure different points on a tilted object.

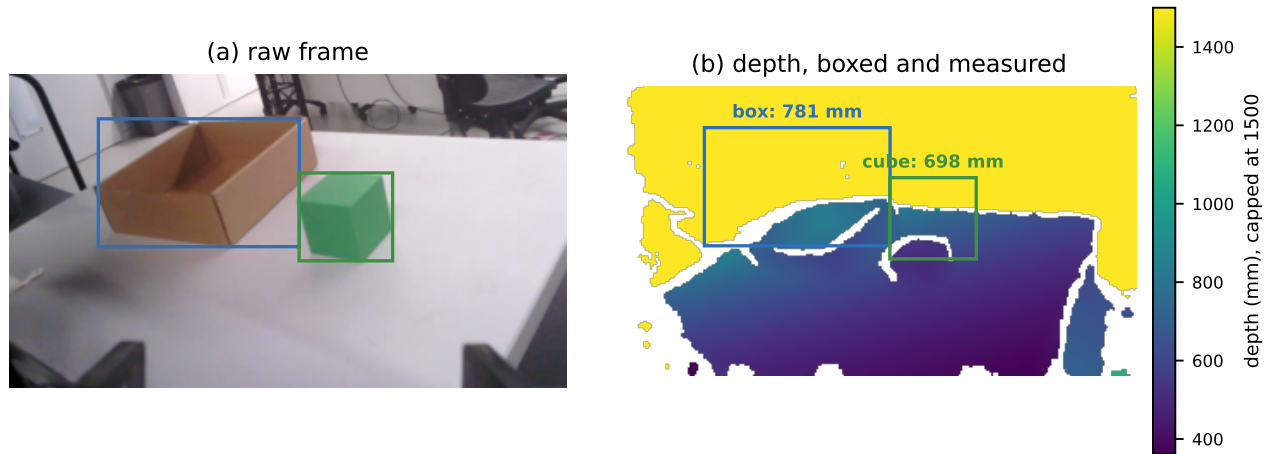


Figure 4.3: Figure 4.2’s frame (a) beside its depth map (b), boxed and labelled with each object’s median measured range.

No grasp closed on any camera this session: the nearest object’s narrowest axis measured 144 mm against an 85 mm jaw, a statement about the scene rather than the planner. Figure F.1 places this against the safety layer’s own subject in one frame: wearer, table, the wearable arms and the objects between them.

In simulation, the full task set solves 13 927 inverse-kinematics calls with zero failures and zero clearance breaches. Sweeping the shared-autonomy intent estimator against a mis-declared table shows why: from declared coordinates it becomes *confidently wrong* 50 % of the time once the declaration is one object-pitch out; from measured coordinates it stays at 0.0 % across the same sweep.

Table 4.2: Every value from the one real frame of [Figs. 4.2](#) and [4.3](#).

Quantity	Value	Method
Support-plane RMS	1.43 mm	RANSAC, this frame
Support-plane inlier fraction	74.0 %	RANSAC, this frame
Cube range	525.5 mm	plane-fit centre
Cube range	698 mm	depth-box median
Box footprint	341×278 mm	oriented 3D box
Box range	677.6 mm	plane-fit centre
Box range	781 mm	depth-box median

4.4 A pilot of direct control against shared autonomy

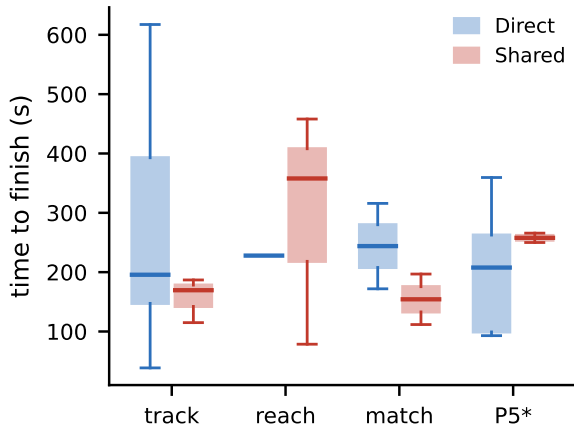
Five operators (P1–P5) completed twenty-three recorded sessions under the protocol of [Section 3.5](#), each on one of three VR-teleoperated tasks – *object tracking* (following a moving held object with the end-effector), *target reaching* (a discrete reach to a marked position) and *position matching* (driving to and holding a target pose) – once under direct control and once with shared-autonomy assistance. P5’s seven sessions predate the per-task naming convention and are pooled as one unlabelled block, identified as direct/shared from the session’s own logged flag. A fourth, planned task using a multimeter as the manipulated object was attempted by the master-arm cohort below but never recorded on any of its three tries ([Chapter L](#)). Coverage is uneven by design – not every operator performed every task under both conditions in the time available – so the pooled comparison below mixes tasks together and the paired comparison that follows it is the part of the data that isolates the condition cleanly. Full session-by-session data, the coverage table, and the supplementary measures are in [Chapter L](#).

Table 4.3: Twenty-three pilot sessions, pooled by condition. Duration includes pausing and hesitation, a session-length rather than a motor-execution measure. E-stop holds ([Section 4.1](#)) are excluded here as engagement time, not a safety verdict.

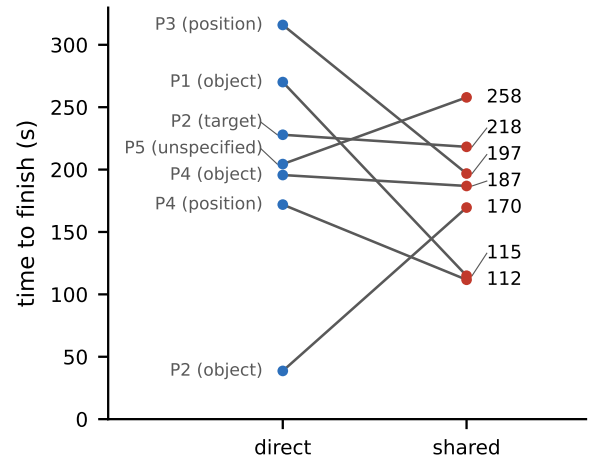
Measure	Direct ($n = 13$)	Shared autonomy ($n = 10$)
Time to finish, mean (s)	241	219
Hand travel distance, mean (m)	2.30	2.78
Re-grips per session, mean	25.8	23.3
Share of session actively steering, mean	55 %	61 %
Robot’s lag behind the operator’s hand, mean (mm)	2.6	2.8

Pooled across all twenty-three sessions ([Table 4.3](#), [Fig. 4.4a](#)), shared-autonomy sessions averaged shorter completion times (219 s against 241 s) and fewer re-grips per session (23.3 against 25.8), with a robot lag under 2.9 mm in both conditions – but *more* hand-travel distance (2.78 m against 2.30 m), a direction that holds in the paired comparison too ([Section 5.4](#)), not just the pooled one ([Fig. 4.5](#)). [Figure 4.4b](#) isolates the seven cases in which the *same* operator repeated a task under both conditions: five of seven finish faster under shared autonomy, and six of seven travel further doing it.

This pilot measures operator performance only, at pilot scale, with no evidence of a safety or responsiveness cost to shared autonomy. It has no separate wearer and so cannot speak to the wearer-side hypotheses that motivated [Chapter J](#)’s two-person design; [Section 5.4](#) treats extending it.

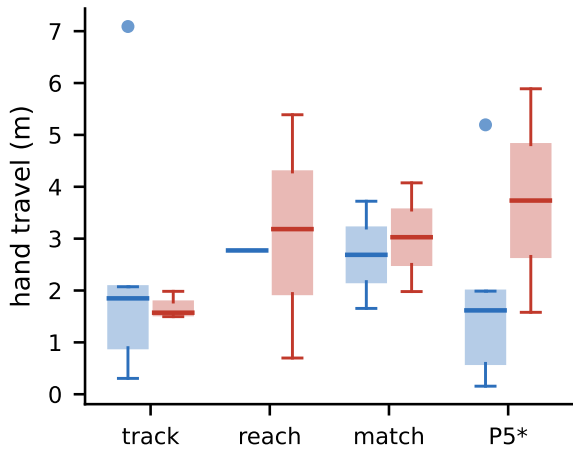


(a) pooled by task (P5's task was unrecorded, *P5**)

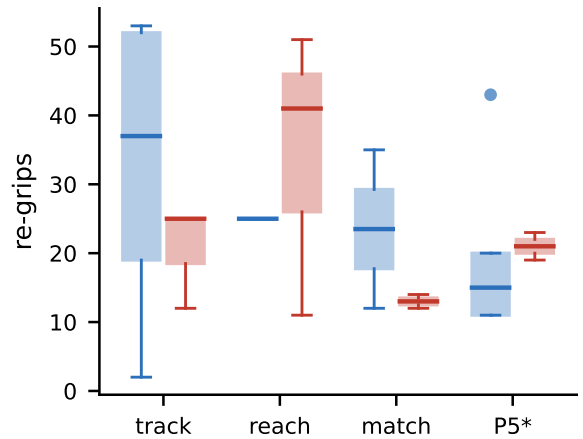


(b) paired within operator, both conditions

Figure 4.4: Time to finish, pooled as box plots – median, quartiles, whiskers and outliers, not just a mean (a) – and paired (b, the fairest comparison here). Duration includes setup and hesitation, so it is the least precise pilot measure.



(a) hand travel



(b) re-grips

Figure 4.5: The counter-intuitive half of Table 4.3: shared autonomy's own box sits *above* direct's in hand travel for three of four task groups, not below, while re-grips move the other way.

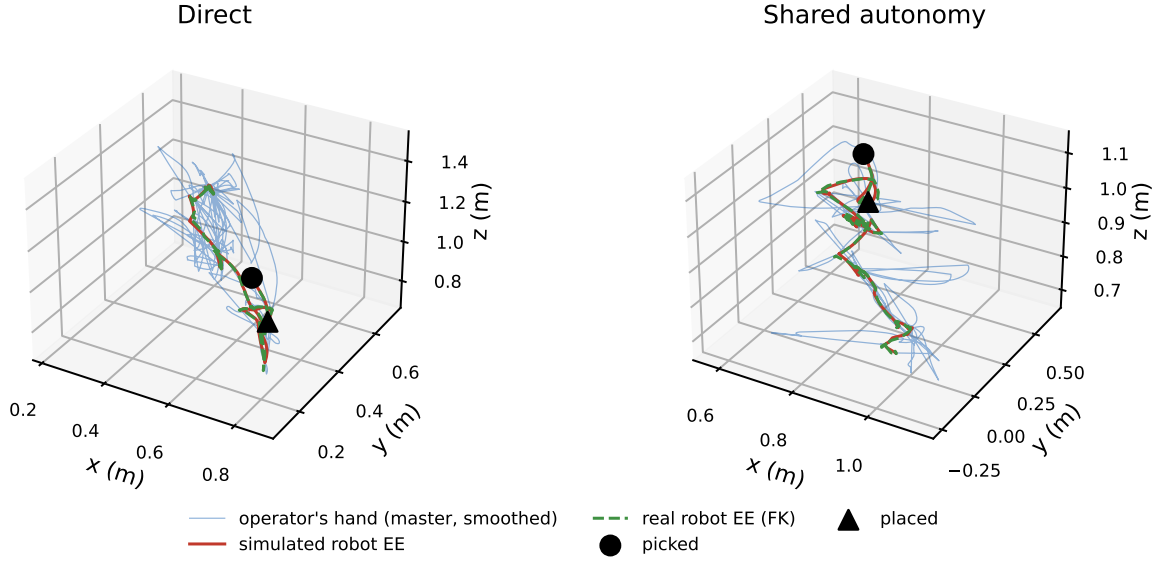


Figure 4.6: VR cohort: master hand (smoothed, the raw input), simulated EE, and the real robot's own EE (forward kinematics on its own encoders) – one Direct and one Shared session, same operator and task, picked/placed from the gripper's logged transitions. Simulated and real agree closely, the same small gap as Fig. 4.8.

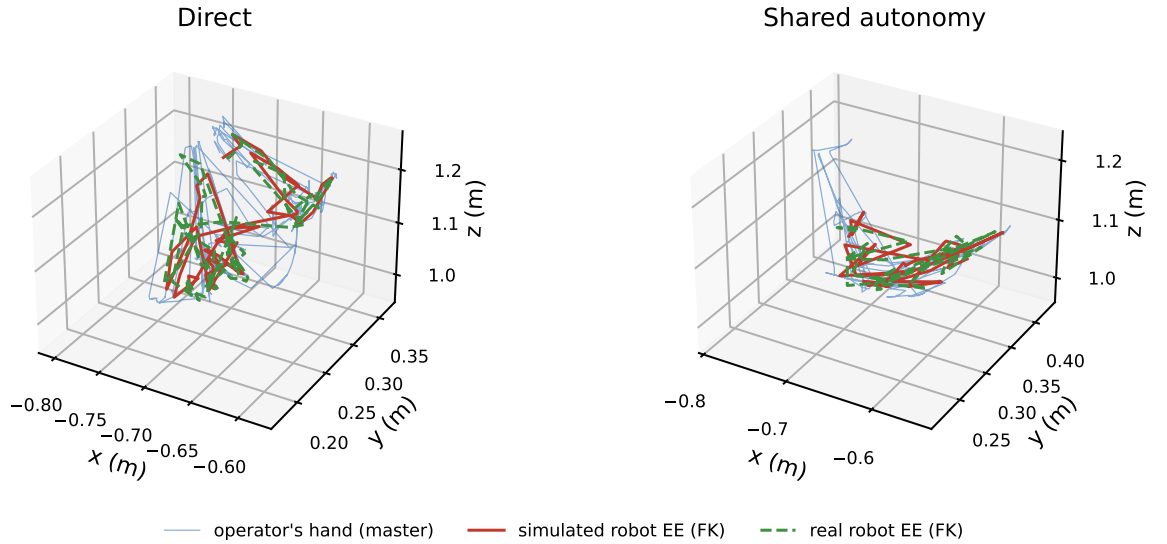
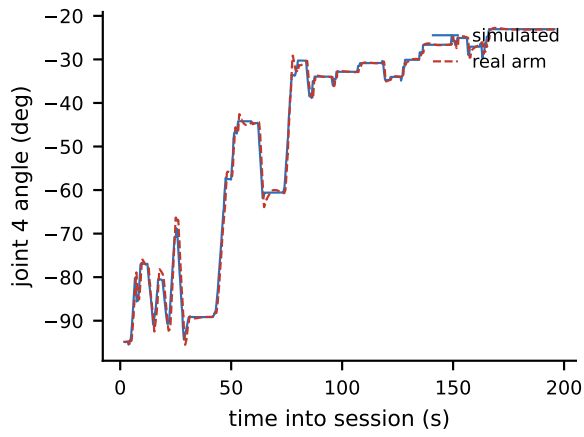
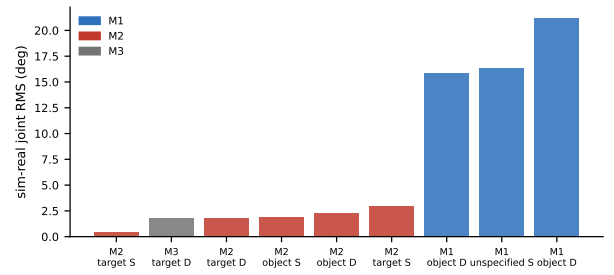


Figure 4.7: The master-arm cohort's own version of Fig. 4.6: M2, Direct and Shared, same task, both under 2.3° sim-to-real RMS. This cohort logs no gripper channel, so no picked/placed markers are drawn.



(a) one joint, best VR session



(b) master-arm cohort, every session

Figure 4.8: Sim-to-real tracking, the same pilot. (a) One joint tracking its command to 1.06° RMS. (b) Three further operators (M1–M3) from the mannequin master: two track within 3° ; the third’s three sessions track 15.9 to 21.2° , consistent with [Section H.2](#). Full record in [Chapter L](#).

Chapter 5

Discussion

5.1 The workspace result, and what it costs

The disjoint-workspace finding is severe because it is geometric: no planner or orientation policy can produce a point neither arm can reach, so inter-arm handover was excluded from the task set before anything else excluded it. The cause is not the mount’s rotation but the arms’ asymmetric home joint angles, ground truth because the sim-to-hardware bridge replays them onto the real arm; fixing it needed a mechanical change, costed, applied and re-verified as one deliberate pass because every clearance and task coordinate in the report depends on it.

5.2 The safety layer is thorough about distance and silent about time

The clearance floor is a *distance*, and the wearer can move. Combining the measured wearer-tracking latency with literature limb-speed ranges gives [Table 5.1](#): in most modelled limb motions the limb travels further than the entire 150 mm margin before the perception chain is even aware anything changed – not an implementation defect, but the consequence of a fixed margin meeting a perception chain of roughly half a second and a person who can move a metre in that time. It is this report’s principal open safety question.

Table 5.1: Limb travel during the measured wearer-tracking latency against the 150 mm clearance floor; full table and six limb-motion classes in [Chapter E](#).

Latency	Deliberate reach	Quick reach or startle
62 ms (median)	62 % of the floor	exceeds the floor
562 ms (worst)	exceeds the floor	exceeds it several times over

5.3 Validity of the measurements

Twenty-six times in this project a measurement was wrong and the system was not ([Chapter G](#)), a high enough base rate to condition how every other number here should be read. Fifteen of the twenty-six were caught by an independently known quantity contradicting the reading rather than by inspection – the argument for building a cross-check into every instrument rather than trusting it harder. Every result above except the pilot and channel measurements comes from simulation with mock hardware, whose force channels are identically zero, so any force-derived quantity is structurally undefined; and every latency reported is control-path latency between software stages, excluding the operator, display and physical arm, so it is a floor.

5.4 What the pilot shows, and its limits

The pilot is the first evidence from real operators on this platform, and it points the same direction as the shared-autonomy literature developed for robots not attached to anybody [DS13, JSB15]: shorter completion times and fewer re-grips under assistance, with no measurable cost in responsiveness. It should not be read as more than that. Hand-travel distance is *higher* under shared autonomy, and unlike completion time this holds in the paired comparison too: six of seven paired cases travel further under assistance, not less. This pilot cannot say why – plausibly the arbiter nudges a directed hand along a route that is not the shortest the operator would have taken alone, making “shorter but less direct” a real property of it rather than a contradiction, but seven paired trials cannot separate that from task-order or fatigue. Five operators with uneven task coverage is a pilot-scale sample, not a powered comparison, and the pooled figures mix three tasks (plus P5’s unlabelled sessions) together; the paired comparison isolates the condition, and rests on seven cases. Session duration includes setup and hesitation, which is why hand travel and re-grips – drawn from the position trace, not a stopwatch – are read as the more reliable measures. Most importantly, the pilot has no separate wearer: it measures what the operator experienced, not what carrying 17 kg of assisted motion costs a second person, the comparison [Chapter J](#) remains the target for.

5.5 Against the planning report

The planning report specified physical springs at the master and virtual impedance at the slave. Neither survives contact with the measured system: the arm’s cyclic interface is unusable over the network path, so commands go through a high-level interface at 18.4 to 18.7 Hz, and an impedance law at that rate is not one. The compliance sought would need to come from the mechanism – series elastic actuation at the joint [TGT⁺22, LKK19] or a current-based formulation avoiding a fast torque loop [dWKB24] – both hardware changes. The safety architecture, degradation architecture, workspace characterisation and pilot were not in the plan; each followed from a measurement the plan did not anticipate.

Chapter 6

Conclusion

This project built a teleoperation system for two seven-degree-of-freedom arms mounted on a worn backpack frame, commanded by a second person through an instrumented mannequin master or VR controllers, with a safety architecture appropriate to a manipulator standing beside somebody who is not commanding it and a shared-autonomy layer above it – and ran a first pilot comparison of direct and assisted control with real operators.

6.1 Against the objectives

A master was built and characterised: six of fourteen channels alive, the repair identified to two specific joints. The command mapping degrades predictably – vertical and fore-aft track correctly, lateral does not, and no recalibration fixes it. The safety layer handled fourteen of fourteen injected faults with every blocker observable by name. The workspace characterisation found the two arms share no reachable point under the as-built mount, identified a buildable mount correction, and that correction was applied and re-verified. Ten-repeat, known-answer verification ran throughout, catching twenty-six measurement defects before they entered this report as findings. And a first pilot ran under ethical approval with five operators across three tasks, showing shorter completion times and fewer re-grips under assistance, at the cost of more hand travel per session and with no measured cost in responsiveness.

6.2 What was not achieved

No result here was validated on a physical Kinova arm carrying a load. The full two-person study – operator and wearer, measured together under the same autonomy – was not run; the pilot substitutes a smaller, operator-only comparison, made possible once VR controllers offered a route around the master’s damaged channels. The planning report’s haptic objective was not delivered, and is not deliverable through this platform’s control interface without a hardware change.

6.3 Further work

In priority order: repairing the two identified master channels, so the mannequin master becomes a usable direct-control baseline; calibrating the wrist-camera-to-arm transform, the largest unmeasured budget term and the reason no open-loop grasp has closed on hardware; correcting the clearance check to sample link geometry rather than link origins – the one item here that is a safety defect rather than an incompleteness; and running the full two-person protocol of [Chapter J](#) with a separate wearer, now that approval and a working input path both

exist, to measure what this project was ultimately built to ask: the exchange rate between what shared autonomy buys an operator and what it costs the person carrying the arms.

6.4 Closing

The most transferable outcome of this work is not the mapping, the safety architecture, or the pilot's result. It is the discipline that produced them: twenty-six times a result that looked like a system failure turned out to be a defect in the measuring tool, and the habit of establishing which one is in hand before acting on it is the part of this project most worth carrying into the study it was built to enable.

Appendix A

The master arm

The master is a miniature mannequin reproducing the Gen3's joint sequence at roughly a third of its scale, moved with the fingers of one or both hands. Source: `firmware/master_arm/teensy_final.ino`.

A.1 Mechanical chain and electronics

Each arm carries seven potentiometer-instrumented joints, alternating a roll about the preceding link's own axis with a bend perpendicular to it; link lengths sum to 0.272 m. A roll joint moves the tip only through a downstream bend (rotating a straight arm about its own axis translates the tip by zero), which is why the loss of J2 or J4 costs reach while J3 or J5 costs almost nothing.

Master arm, from CAD — zero configuration

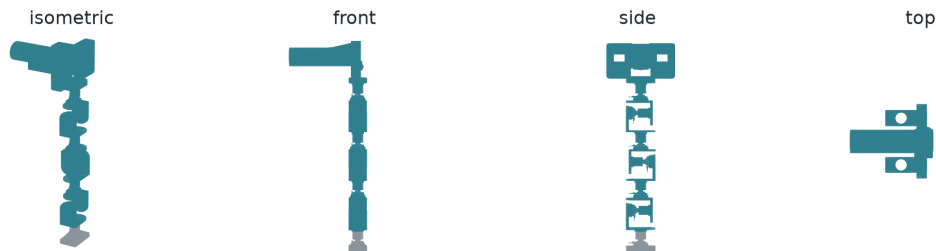


Figure A.1: Four orthographic views of the assembly at the zero configuration. The mechanical model carries seven potentiometer housings, a shaft coupling at each joint, and a wrist platform for the inertial sensor and force-sensitive resistor.

A Teensy 4.1 reads fourteen potentiometers, two MPU-6050 inertial sensors, two force-sensitive resistors and two buttons, emitting one ASCII text frame per cycle at 100 Hz ([Table A.1](#)). An earlier firmware revision placed two potentiometer channels on A4/A5, which are physically the default Wire bus both inertial sensors use; calling `analogRead()` on a pin the I²C peripheral is driving corrupts the bus intermittently and returns meaningless potentiometer values, with no symptom that looks like a pin conflict from the outside. The fix excludes both pins from analog use: Teensy 4.1 exposes eighteen analog-capable pins, minus two leaves sixteen, exactly what fourteen potentiometers plus two force sensors require – no spare.

Table A.1: Serial frame, in transmission order.

Field	Count	Format	Meaning
k1j1..k1j7	7	%.1f	left arm joints, 0 to 360°
k2j1..k2j7	7	%.1f	right arm joints
K1IMU, K2IMU	6+6	%d	$a_x a_y a_z g_x g_y g_z$, raw LSB
fsr1, fsr2	2	%d	grip force, raw counts
btn1, btn2	2	%d	latched toggle state

A.2 Signal conditioning

Each potentiometer channel is read at 12-bit resolution and passed through an integer exponential moving average,

$$e_{n+1} = \frac{\alpha r_n + (256 - \alpha)e_n}{256}, \quad \alpha = 64, \quad (\text{A.1})$$

a smoothing factor of 0.25 per sample. The filtered count maps to degrees by a clamped linear transform that discards the outer 50 counts at each end of the ADC range, where a potentiometer wiper is least linear.

Recorded captures show each potentiometer producing a new *distinct* value at only 14.7 to 17 Hz against the 100 Hz frame rate, so 82 to 88 % of consecutive rows repeat – every per-row statistic computed before this was noticed was invalidated by it ([Chapter G](#)). At the power-on default of ± 2 g full scale, accelerometer sensitivity is 16 384 LSB/g; the widely copied value of 8192 LSB/g corresponds to the ± 4 g range and, used here, halves every reported acceleration – invisible in the elevation estimate (a uniform scale error cancels in a normalised vector) but fatal to the accelerometer gate, which then rejects every frame and looks like a dead sensor. Force-sensitive resistors are read as raw integers with no filtering; the mapping applies its own deadband and hysteresis downstream, so the discrepancy between the design (which specifies a filtered value) and the firmware in hand is recorded rather than resolved.

A.3 The CAD is not the kinematic source

Neither STEP file contains a single kinematic-joint entity; they are solid geometry for manufacture, describing one fixed configuration. No URDF or xacro anywhere in the workspace references either file – **the master arm has no URDF at all**. What the software uses is seven measured constants in `srl_teleop/master_calibration.py`, taken directly from the physical arm. The CAD’s joint-module pitch agrees with those constants to 1.2 %, which is a cross-check, not a licence to treat the CAD as the source: it cannot distinguish the 43 mm roll from the 37 mm bend within one measured pitch.

Appendix B

The route the design took

A record of process, not of result: the routes tried and abandoned before [Chapter C](#) took its delivered form. Recovered prior programs are archived at `thesis_report/_source/prior_work/`, dated 7 June to 27 July 2026; the full divergence analysis against the planning report is at `thesis_report/DIVERGENCE.md`.

The first master-to-slave mapping sent master joint angles directly to slave joint angles. It failed for a structural reason rather than a tuning one: forward kinematics compounds every joint angle in the chain, so a 9 mm/3° disagreement between the master’s CAD and its measured link lengths ([Section A.3](#)) propagates to a materially wrong slave pose at the wrist, and there is no reconstruction path once a channel carrying an intermediate joint fails. Task-space mapping, sending master *tip displacement* to slave end-effector displacement, decouples the two chains and is what shipped – at the cost of leaving the slave’s redundant degree of freedom unspecified, which is what [Chapter I](#) later shows to be expensive near the wearer.

Two further redirections are worth recording because each was forced by a measurement rather than chosen for elegance. The planning report’s haptic objective – physical springs at the master, virtual impedance at the slave – was abandoned once the arm’s cyclic control interface was found unusable over the available network path ([Chapter 4](#)): an impedance law running at 18 Hz is not an impedance law. And validating the master-slave mapping in simulation before touching hardware, as planned, used MuJoCo [[TET12](#)] rather than Isaac Lab [[NVI26](#)] – it starts in under a second, its model format is one readable file, and for a kinematic-mapping question its contact solver is irrelevant, so the heavier alternative bought nothing.

Sensor faults turned out to be the dominant failure mode, not control faults: injecting a stuck analog reading sent a wrong angle to the corresponding slave joint with no reconstruction available under the joint-space mapping, which is the finding that motivated the graceful degradation architecture of [Chapter C](#) and, with it, the reduced azimuth/elevation/reach decomposition that the task-space mapping made possible.

Appendix C

System as built

Chapter 3 gives the command mapping and safety architecture in summary, including the package architecture (Fig. 3.1). This appendix covers what is not repeated there: the full mode registry, the desk arrangement the laboratory actually operates, calibration, the operating window, and the verification audit.

C.1 Command mapping, in full

The commanded position is built in spherical form about the master’s shoulder rather than by forward kinematics from seven joint angles, each component taken from whichever sensor observes it most directly:

$$\phi = \arcsin\left(\hat{a} \cdot \frac{\mathbf{g}}{\|\mathbf{g}\|}\right) \quad \text{elevation, from gravity} \quad (\text{C.1})$$

$$\psi = \theta_1 - \theta_1^0 \quad \text{azimuth, from the shoulder roll} \quad (\text{C.2})$$

$$r = \|\text{FK}(\theta_1, \theta_2, \theta_3, \theta_4, 0, 0, 0)\| \quad \text{radius, magnitude only} \quad (\text{C.3})$$

with $\hat{a}$ the arm’s long axis in the sensor’s own frame, captured once at rest, and $\mathbf{g}$ the measured acceleration. Elevation is mount-independent and drift-free; radius is dominated by the bend joints and tolerates a dead roll passed through as zero. The commanded end-effector position is

$$\mathbf{p} = \mathbf{p}_{\text{anchor}} + s(\mathbf{d} - \mathbf{d}_{\text{ref}}), \quad (\text{C.4})$$

with $\mathbf{d}$ the master tip in the robot frame and $\mathbf{d}_{\text{ref}}$ the reference latched at clutch engagement, from the first *valid* frame after boot at the master’s actual pose. An earlier version initialised the reference to the origin of the master’s own frame, so the first command was a step of the master’s entire tip vector (roughly 0.22 m) in whatever direction the master happened to be lying.

C.1.1 Collision avoidance in three stages

The wearer is in the collision model: thirteen collision elements were added, paired with thirty-six exclusions for the mount-adjacent proximal links (the arm bases are bolted to the harness and permanently adjacent, so without exclusions every pose is trivially in collision – everything distal of the first arm link stays enabled, which is what actually protects the wearer). Avoidance is graduated: collision-aware inverse kinematics rejects a configuration; on rejection the redundant degree of freedom is re-seeded and the solve retried; a geometric clearance floor measured from the transform tree provides the final margin. The floor is measured against the *fused* body of

Section D.4, not the semantic collision description, which excludes forty-four proximal pairs and would report a pose valid with the upper-arm tube inside the person; and `base_link` sits 0.1610 m from the torso and is moved by no joint, so a clearance reading of exactly that value is a measurement of the mount, not the arm (Chapter I). Driving the commanded pose directly at the wearer’s head and torso stopped the arm at 0.072 to 0.079 m with zero clearance-floor blocks, so the earlier stages did the work and the floor remained a backstop.

C.2 Control modes, in full

Table C.1: Every delivered operating mode. The numbering is 01, 02, 03, 04 and 06: there is no 05, and the gap is preserved because recorded data carries these labels.

Mode	Input	Robot decides	$v_{\max}$ (rad/s)
01 direct, mannequin	master arm	nothing	0.60
02 direct, controllers	tracked controllers	nothing	0.60
03 shared autonomy	master, vision	wrist orientation	0.60
04 shared autonomy, controllers	controllers, vision	wrist orientation	0.60
06 full autonomy	typed or spoken goal	target, approach, grasp, place	0.15

Orientation assist is a stub: it would supply wrist orientation under an orientation constraint, and the inverse-kinematics plugin in use ignores the constraint field entirely (270 solver calls, success identical with and without it). It needs a constraint-aware plugin or explicit yaw sampling.

C.2.1 The desk arrangement

The laboratory’s actual arrangement seats the operator across the room, facing the wearer, holding the VR controllers as a six-degree-of-freedom input device and watching the physical robot directly. **Nobody wears the headset**; it stands on a shelf as the tracking reference for the controllers. The alignment between the operator’s frame and the robot’s is a measured yaw and can never be a mirror: facing someone and copying their motion is a reflection (determinant -1), which looks correct in position alone but reverses every commanded orientation. The angle is measured by having the operator move 0.4 m toward the robot, with a perpendicular cross-check; calibration refuses to write a value if the two motions are not perpendicular and horizontal.

Bringing the WebXR transport up for the first time (2026-08-19) exposed two defects no unit test would have found: the bridge node had never started at all, because an attribute named `clients` shadowed a read-only property of the base class; and its axis map had been written for the conventional robotics frame rather than this project’s, so a hand moved 3 m forward commanded 3 m to the wearer’s right. The delivered route serves the client page and the wss socket from one origin so a single certificate exception covers both, needs no adb or sideloading, and was verified end to end on 2026-08-19.

C.3 Calibration

C.4 The operating window

Every capability reaches one window: a feature reachable only from a terminal does not exist for the person running a session with a wearer standing in the rig. `scripts/verify_gui_buttons.py` presses every control and checks what the press did (257 checks). The first complete run found four defects: five session controls

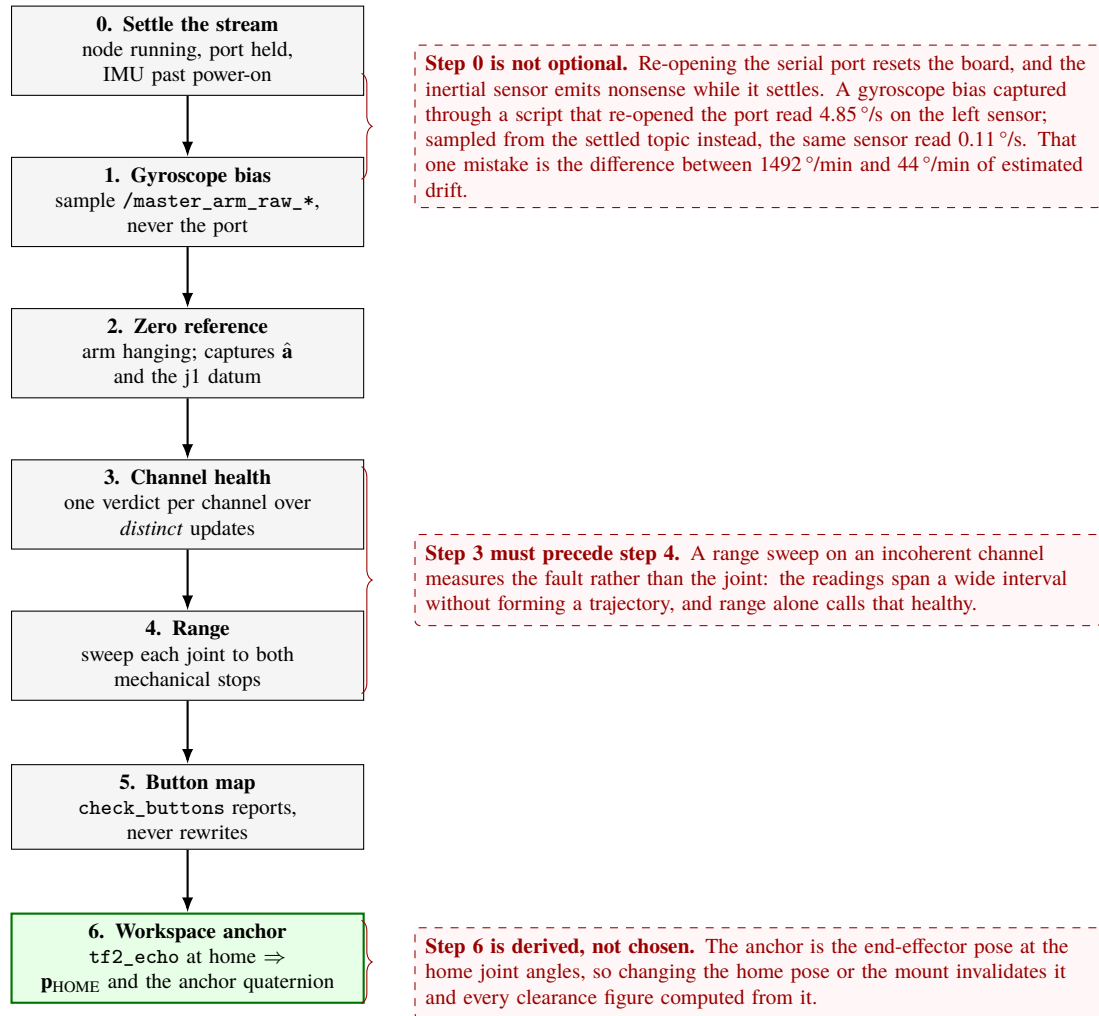


Figure C.1: The calibration procedure, in order. A calibration routine measures whatever the instrument was actually doing at the time, so establishing that the instrument is in a valid state is part of the measurement and not a preliminary to it.

raised an exception inside a Qt slot, which the framework swallows, so they did nothing and said nothing; and the stray-process preflight for the one-stack constraint had never executed, because its module was never imported.

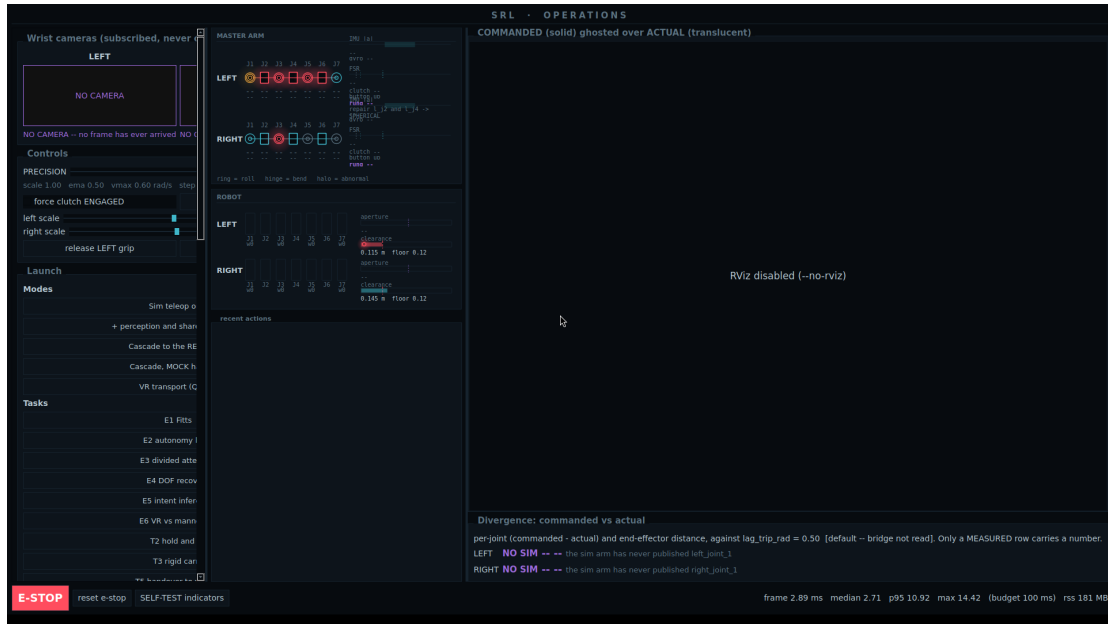


Figure C.2: One command opens it. Left: set-up, run, status. Centre: mode registry and per-mode controls. Right: the three-dimensional commanded view beside a panel drawn from the arms' own reported joint states, with the divergence between them read out beneath both, so a mismatch between commanded and reported state is visible without reading a log.

C.5 Verification methodology

The rule – a surprising failure is evidence about the instrument until the instrument has been cleared – and the ten-repeats-over-a-densified-path practice are stated in [Chapter 3](#); [Chapter G](#) is the full case-by-case record.

Appendix D

Perception, environment mapping and autonomous manipulation

Chapter 3 summarises this stage. Almost everything in the delivered task layer previously ran on *declared* coordinates verified at ten repeats over densified paths, none of which survives a different table, a different object, or the same object displaced by 50 mm. Three components capable of sensing the same quantities already existed and had never been composed: a support-plane finder used only to print a description, a voxel-world planner constructed nowhere outside its own self-test, and a work-surface module whose depth-driven height setter nothing called.

D.1 Detection and grasp synthesis

Detection is open-vocabulary (YOLO-World [CSG⁺24] with FastSAM [ZDA⁺23]), falling back to a hue-and-depth filter; the backend that produced a detection is reported with it. Range is used as a discriminator where hue cannot be: the laboratory’s teal robots and the target green block differ in hue by only two units, well inside noise, but sit 2.8 m away against a 1.6 m working range, so range separates them at zero extra cost – the same size-at-range expression ($w = sZ/f$) also rejects a 210 mm pad presented as a 40 mm cube. Grasp synthesis takes the principal axes of the object’s depth cloud: writing the mean-centred points as $\mathbf{P}_k$, the axes are the eigenvectors of the scatter matrix

$$\mathbf{C} = \frac{1}{N} \sum_{k=1}^N (\mathbf{P}_k - \bar{\mathbf{P}})(\mathbf{P}_k - \bar{\mathbf{P}})^\top, \quad \mathbf{C}\mathbf{u}_j = \lambda_j \mathbf{u}_j, \quad \lambda_1 \geq \lambda_2 \geq \lambda_3, \quad (\text{D.1})$$

and the jaw closes across $\mathbf{u}_3$, the direction of least extent, capped at

$$w = \max_k \mathbf{u}_3^\top \mathbf{P}_k - \min_k \mathbf{u}_3^\top \mathbf{P}_k \leq w_{\max} = 85 \text{ mm}. \quad (\text{D.2})$$

A geometric method can *name* the cause of a refusal (too wide, no depth, too few points), which a learned alternative cannot and which also would not fit beside the detector in 4 GiB of video memory.

A reachability gate checking candidate grasps for reach and wearer clearance was inoperative for the life of the module: it scored the clearance from the whole kinematic chain including the immobile mount stub, which reads a constant 0.2202 m in every pose and so could never fail against the 0.15 m floor. It now reads the moving-chain clearance (0.4031 m at home) and a regression test requires a named refusal when aimed at the wearer’s torso.

D.2 Measuring the working area

The sweep holds one wrist attitude per pass and translates, printer-bed-probe style, rather than re-aiming at each grid point: re-aiming would make a residual camera-to-wrist error a *different* error in every frame, which can neither cancel across views nor be estimated from them. A frame is accepted only once no joint has moved by more than 0.0015 rad over 400 ms – a tenth of the arrival tolerance.

Table D.1: The fused map after one full sweep, against the scene it was measuring.

Quantity	Measured
Surface height	1.250 26 m against a declared 1.250 m
Objects recovered	6 of 6, every centre within 1 mm
Object extents	every dimension within 3 mm
Coverage of reachable cells	98.65 %, 0 stillness refusals
Elapsed	13 min 15 s

An earlier surface error of -5.2 mm was itself an instrument defect ([Chapter G](#)): the reported height was the fitted plane’s intercept at the *origin*, a point inside the wearer, not its height over the working area. Reachability (a fact about the robot, computed once) and the scene map (a fact about the room, rebuilt from scratch every time) are separate products with separate lifetimes; `scripts/pick_from_map.py` refuses to move from a map older than thirty minutes, because a stale map is indistinguishable from a fresh one by inspection.

D.3 Planning and natural language

`joint_planner.py` implements RRT-Connect [[KL00](#)] in joint space (the binding constraint is the whole arm’s distance to the wearer, not the hand), evaluated against the same moving-chain clearance the guard enforces, with every edge densified at 0.05 rad to match the path checker. Without a planner, a pick whose pre-grasp and grasp poses were each individually clear passed within 0.1462 m of the wearer mid-path.

Table D.2: Seventy-five phrasings, delivered grammar against an earlier revision checked out of version control.

Outcome	Earlier	Delivered
Correct	21	34
Asked a clarifying question	14	14
Refused	37	27
Misunderstood	3	0

The three earlier misunderstandings shared one mechanism: a compound instruction (*put the blue ones on the blue mat and the green ones on the green mat*) was cut at the first destination cue, so the two-target check counted targets only in the head clause, planned the first, and silently discarded the second while reporting success.

D.4 Observing the wearer

The clearance floor of [Chapter 3](#) is enforced against a body. `wearer_tracker_node.py` replaces the mannequin in the robot description, part by part, with one built from markerless pose estimation and metric landmarks lifted by perspective-n-point [[LTN⁺19](#)]: 28 to 34 ms inference, 61.6 ms median frame-to-estimate, 62 to 562 ms end to

end (the spread is a configured publication rate limit, not the estimator). The safety argument is one monotonicity rule enforced in one function: **the camera may only ever make the wearer larger** – per body part, `fuse()` keeps whichever of the tracked and mannequin primitive is *nearer* the robot. A body injected 0.10, 0.30, 1.00, 5.00 m further away moved the enforced margin by nothing; six injected faults (dark frame, blank frame, two people, empty frame, occluded arms, a control) were refused seven of seven, each naming the fault. The honest limitation is on the fallback side: if the wearer raises their arms and tracking is then lost, the guard reverts to arms-down and believes 0.317 m of clearance where there is 0.093 m.

Before this was connected, the tracker reached the planning scene and *not* `mount_guard_node`, which builds its wearer at import time – the body the planner avoided and the body the floor enforced against were two different bodies. A second camera observing the workspace from outside the arms is implemented, refuses to republish stale frames, and both calibration tools pass self-tests to 0.43 % focal length and 0.0000 m/0.000° extrinsic error – but **no camera has ever been attached to it**: the device is the host’s own webcam, and forwarding it into the Linux environment needs an administrator-level bind that was refused.

Appendix E

Motion generation and simulation-to-hardware transfer

Chapter 4 reports the headline numbers. This appendix gives the mechanism.

E.1 Why a per-joint clamp cannot stay on the line

The limiter it replaced applied, per cycle and independently to each joint i , $q_i^{(k+1)} = q_i^{(k)} + \text{clip}(q_i^* - q_i^{(k)}, -\Delta, +\Delta)$ with one step bound Δ shared by all seven joints. Writing the required displacement as $\mathbf{e} = \mathbf{q}^* - \mathbf{q}^{(k)}$, staying on the straight joint-space line requires the applied step be *parallel* to $\mathbf{e}$; the clamp applies $\text{clip}(\mathbf{e}, -\Delta, +\Delta)$ component-wise, which is parallel to $\mathbf{e}$ only when every component is inside the bound or every component saturates equally. In every other case the direction of travel rotates and the end effector leaves the path. This is why sweeping Δ over a factor of 35 does not drive the excursion to zero – it settles at 68 mm – and why the excursion is a property of the algorithm, not its tuning.

E.2 What replaced it

Ruckig [BK21]: time-optimal, jerk-limited, per-degree-of-freedom limits, accepting a non-zero target velocity. Phase synchronisation drives every joint from one scalar phase $\sigma(t) \in [0, 1]$:

$$\mathbf{q}(t) = \mathbf{q}^{(k)} + \sigma(t)(\mathbf{q}^* - \mathbf{q}^{(k)}), \quad \sigma(0) = 0, \sigma(T) = 1, \quad (\text{E.1})$$

so the commanded configuration is on the segment between endpoints by construction for every t – unlike merely stretching each joint’s own profile to the slowest joint’s duration (time synchronisation), which measures 9.5 mm of path deviation on the same case against phase’s 0.02 mm. The acceleration and jerk limits are *assumed* and labelled so: `joint_limits.yaml` declares zero acceleration limits for all fourteen joints, so there is nothing to read; only the velocity limits are the robot’s own.

Five defects were caught in the replacement, four of which would have passed a conventional test suite: re-planning every cycle destroyed phase synchronisation (a mid-profile re-plan cannot reproduce the profile it is half through); floating-point noise forced a re-plan every cycle even after fixing that, because the target was recomputed via a wrapped subtraction not bit-identical to itself; the synchronised-clamp fallback reported zero acceleration (because it does not model one) and so passed a continuity check by fabricating the quantity being checked; the off-path metric had a 0.1 mm grid-resolution floor and was reporting the instrument’s resolution as

the robot’s behaviour; and an arrival test used an absolute tolerance on travels spanning a factor of twenty-five.

E.3 The simulation-to-hardware gap

Of 252 recorded per-joint errors, 211 lie between 0.25° and 0.30° with the sign following the direction of travel – a *bound*, not a spread, which identifies a terminal offset rather than noise. Two candidate models, a terminal offset and a proportional gain error, coincide at one travel distance and differ by a factor of five at 20 mm; propagating the recorded errors through the manipulator Jacobian reproduces the measured Cartesian error to 90 % ($r = 0.923$) under the offset model and not the gain model. The data had been recorded for weeks in `recordings/baselines/arm_directional_calibration.json` before anything read it.

An earlier report claimed the terminal error is speed-independent across a fivefold range; the velocity limit was read once at bridge start-up and seven of thirty-six runs moved *faster* than commanded, so the three speeds were three replicates of one condition and the speed dependence has never actually been measured. The bridge deadband was 1.0° against a measured 0.305° park – a factor of 3.3 – and is 0.10° now, with an opt-in overshoot correction; neither has been tried on hardware.

E.4 The positioning error budget

$$\sigma_{\text{RSS}} = \sqrt{\sum_i b_i^2}, \quad \sigma_{\text{worst}} = \sum_i b_i, \quad (\text{E.2})$$

with σ_{worst} the honest figure against the 30 mm capture gate, since the two largest terms are declared constants rather than noise. Corrections – the pad offset re-derived from forward kinematics (13.45 to 0.05 mm), the shell bias removed via the measured support plane, and the terminal-offset overshoot (joint term 7.24 to 1.88 mm) – take the budget from 18.64 to 12.91 mm RSS and 33.39 to 19.99 mm worst case, *provided* the camera-to-wrist transform is small. [Section H.7](#) measured it directly and it is not.

E.5 The reaction budget

Table E.1: Limb travel during the measured wearer-tracking latency against the 150 mm clearance floor. Speeds are literature ranges, labelled as assumptions; latencies are measured.

Limb motion	Latency	Travel	Against the floor
Deliberate reach	62 ms	50 to 93 mm	62 % of it
Deliberate reach	562 ms	450 to 843 mm	exceeds it
Quick reach	62 ms	93 to 155 mm	exceeds it
Startle	62 ms	155 to 248 mm	exceeds it

In five of six modelled limb motions the limb travels further than the entire margin before the guard is aware anything has changed – [Chapter 5](#) treats this as the principal open safety question.

Appendix F

The vision pipeline, measured on the real cameras

Measured on four physical cameras, 2026-08-30 – together with [Section H.7](#), the only hardware numbers in this report.

The pipeline is instrumented at thirty-three stages per camera; every stage draws itself on the frame it was given, states its formula, and reports its quantities with units, so a claim can be checked rather than taken on trust. A stage has three possible outcomes, not two: it succeeds, it *refuses* and names its cause and remedy, or it is *absent by design* (the webcam’s fifteen depth stages, because it has no depth sensor) – rendering an absence identically to a defect has cost this project findings before.

F.1 The orientation defect

Three places in the software independently decided whether the scene depth camera is mounted upside down, and disagreed. Settled by correlating each frame’s grey signature against the HD webcam (rotated by nothing, on the same side of the room) in both orientations: the un-rotated frame agrees at $r = 0.66$ to 0.70 over four captures, the rotated one at $r = 0.27$ to 0.34 – the same verdict every time, a factor of 2.4 apart. **The camera is mounted the right way up**, and every earlier scene figure was inverted. Because the wrong rotation had been applied *consistently* to the image and the intrinsics together, the recovered point cloud was an exact reflection of the true one and every distance, extent and residual was preserved – the defect was invisible to every metric the pipeline itself computes and was found only by comparing one camera’s picture against another’s.

F.2 What the cameras measured

Table F.1: One frame per camera. The inlier fraction decides whether the row below it means anything.

	left wrist	right wrist	scene depth
pixels returning a depth	89.4 %	80.5 %	14.5 %
plane inliers	74.0 %	52.8 %	3.8 %
plane residual, RMS	1.43 mm	1.59 mm	3.41 mm
objects posed in 3D	5	12	12
graspable within the jaw	0	0	0

The wrist cameras measure a work surface to about 1.5 mm – within a factor of three of the same code against a renderer, the first evidence the surface-fitting stage transfers to hardware. **The scene camera does not identify a support plane**: its best plane holds 3.8 % of the points, so every quantity that depends on it (height,

object boxes) rests on a plane one point in twenty-six agrees with. **No object in any frame could be grasped:** the nearest object's narrowest axis measures 144 mm against an 85 mm jaw. **The open-vocabulary detector returned nothing on any camera** for the prompt *green cube*; the classical colour-and-geometry path found the same objects it did not.

Two defects in the first runs were software, not hardware. The pick path's own depth working band, correct for a wrist camera, applied unchanged to a camera across the room retained **0 of 85 061 returns** – the band is now declared per source. And two stages disagreed about the same frame because depth at 480×270 forward-projected into colour at 1280×720 lands one sample in roughly every seven pixels, so the above-plane mask was a dot pattern that failed a connectivity check while a per-mask stage saw it fine; splatting each projected sample raised coverage to 64.4 % and 58.8 %.

The arms were to be parked at the pick pose for this session; measured rather than assumed, the left arm was 10.67° and the right 13.53° from the reference configuration, against a 2° tolerance, so no figure from this session is captioned "at the pick pose."

F.3 The scene the safety layer reasons about

Figure F.1 names four classes in one frame rather than isolated detections: the mannequin the clearance floor of Section 4.1 is enforced against, the table, the two wearable arms, and the task objects between them. No detector in this project covers the first three classes – only the task-object palette does – so all six boxes here are hand-identified against a pixel grid on this exact frame, stated as such rather than presented as automatic detections.

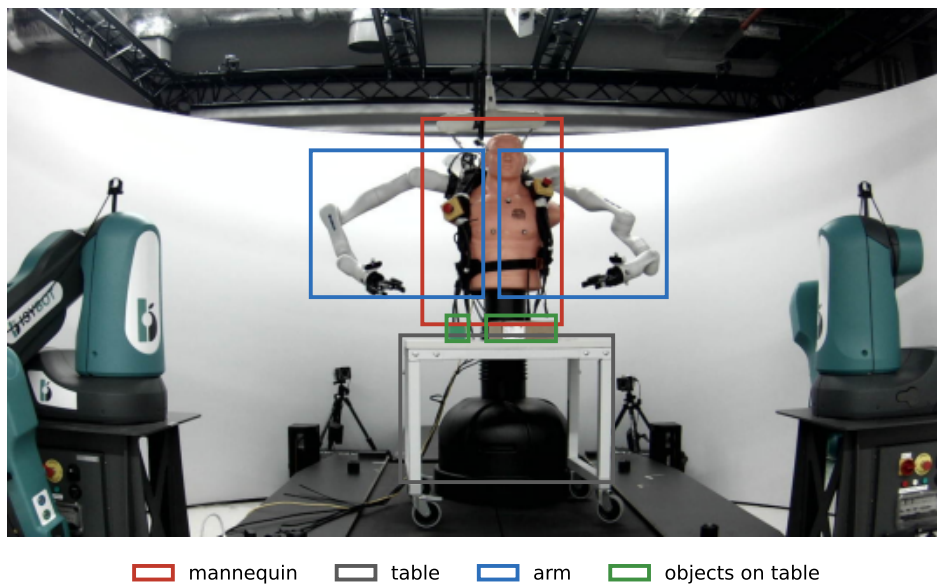


Figure F.1: The same figure referenced from Chapter 4.

Appendix G

Measurement validity: the full record

Chapter 5 states the validity argument and the two numbers that carry it. This appendix is the evidence: what each of the twenty-six cases actually was, grouped by mechanism.

A surprising failure is evidence about the instrument until the instrument has been cleared.

Twenty-six times a measurement in this project was wrong and the system was not. In every case the wrong reading was *plausible* – none announced itself – and two would have removed results now central to this thesis. Sorted by mechanism, they fall into five groups.

An absence read as a value (7 cases). Zero variance from oversampling a slower sensor ([Section H.2](#)); a detection rate of 0 to 4 % from a renderer producing images unlike anything the detector was trained on; a surface height reporting the fitted plane’s intercept at the origin rather than over the working area.

A test harness constructing the condition it tests (6 cases). A reachability control comparing a search-cap lower bound against a boundary, which was never going to pass regardless of the arm’s true reach; a path-verification failure from the verifier’s own test path running in the one direction that leaves the feasible region.

State left over from a previous run (5 cases). A reachability sweep measured with the arms parked 0.87 and 0.59 rad from home, reporting zero failures under the same name as the genuine measurement.

Order of operations inside the measurement (5 cases). An isotropy statistic taken as the minimum over directions of a single sweep, dominated by rare early solver terminations and varying by a factor of two between runs; stable once repeated ten times and read as a median.

A constant asserted in more than one place, measured in none (3 cases). The scene camera’s mounting orientation ([Section F.1](#)); the wrist-to-pad offset, treated as a scalar when it is a function of gripper opening.

Two numbers carry the argument. **Fifteen of the twenty-six were caught by an independently known quantity contradicting the reading** – five by inspecting the code, six by looking at the raw artefact, usually an image. Care was not the defence that worked; redundancy was. **Fourteen of the twenty-six were already answerable from something in hand** – a published stroke, an arithmetic prediction, a quantity exact by construction – which is the recommendation this project’s experience supports: build the cross-check in rather than trying harder.

G.1 What follows from it

Every reachability figure in this report carries its repeat count, because one solver call is one coin flip. Every stage of the simulation campaign carries controls that must *fail*. Where a quantity has never been measured, the text says so rather than offering an estimate – the camera mounting transform in the positioning budget is one such term, and it is not small ([Section H.7](#)). Any unreproduced single measurement in this report should be read as provisional.

G.2 Three standing caveats

Simulation is not the arm. Every quantitative result except the channel measurements and the pilot comes from simulation with mock hardware, whose force channels are identically zero – any quantity derived from force is structurally undefined, not merely unmeasured. **Latency figures are lower bounds**, measured between software stages, excluding the operator, the display and the physical arm’s response. **The one-stack constraint:** two instances of the master node once split the serial stream between them and invalidated a full day of measurements before the cause was found; this is now prevented at the file-descriptor level, but any measurement taken before that prevention carries a risk that cannot be retrospectively excluded.

Appendix H

Results, in full

Chapter 4 reports the headline numbers with their evidence class. This appendix is the supporting record: every table the simulation campaign, the hardware sessions and the safety and instruction-handling measurements produced, with only enough text to say what each is.

H.1 The simulation campaign

`scripts/run_simulation_campaign.py` runs every simulation stage by one command and records what each produced with its exit status and duration; `scripts/make_sim_figures.py` refuses to emit a table whose stage did not run. Sixteen stages, all of which ran; 13 927 inverse-kinematics calls at zero failures.

Table H.1: The simulation campaign inventory.

Stage	Needs	Ran	Time
<i>computer vision</i>			
cv_arithmetic	standalone	✓	12 s
cv_orientation	standalone	✓	< 1 s
cv_replay	standalone	✓	88 s
<i>teleoperation</i>			
teleop_budget	standalone	✓	< 1 s
teleop_motion	standalone	✓	5 s
teleop_orientation_cost	simulated platform	✓	20.6 min
teleop_replay	recorded data	✓	< 1 s
teleop_smoothing	standalone	✓	< 1 s
<i>robot tasks</i>			
tasks_accuracy	standalone	✓	< 1 s
tasks_dance	simulated platform	✓	38 s
tasks_msc	simulated platform	✓	48 s
tasks_status	standalone	✓	< 1 s
tasks_t1	simulated platform	✓	2.2 min
<i>autonomy</i>			
autonomy_arbitration	standalone	✓	4 s
autonomy_instructions	standalone	✓	< 1 s
<i>study design</i>			
study_power	standalone	✓	17 s
<i>transfer to hardware</i>			
sim_to_real	recorded data	✓	< 1 s

H.1.1 Computer vision against a constructed scene

Twenty-one known-answer checks on a frame whose plane, object pose and size were chosen. The fitted plane normal lands 0.0001° from the constructed one; a 40 mm cube measures $40.8 \times 40.7 \times 40.7$ mm and centres to 0.41 mm. Two naive shortcuts are measured wrong by name: an axis-aligned bounding box reads 41 % too wide, and taking the object centre as the mean of its visible points is out by 16.15 mm along the surface normal (a single view sees a shell, not a solid) – corrected to 0.35 mm by assuming the object rests on the fitted plane.

H.1.2 Shared autonomy: what the intent estimator is sensitive to

Sweeping the declared-position error over the range a real table spans, a declared-coordinate estimator becomes *confidently wrong* 50 % of the time once the declaration is one object-pitch out; a measured-coordinate one stays at 0.0 % across the same sweep, with the correct rate flat at 71.7 to 75.7 %. The correct rate never exceeds 76 % in any condition, including perfect information – the remaining quarter is ambiguity, and the arbiter is specified to stay in direct teleoperation whenever ambiguous.

H.1.3 Full autonomy: seventy-five phrasings

Correct interpretations rose from 21 to 34 of 75 against an earlier grammar revision, refusals fell from 37 to 27, misunderstandings fell from 3 to 0. Ten of fourteen clarifying questions carry a scored reply; all ten resolve correctly.

H.2 Channel health

Measured on recorded hardware data, 2026-08-06. Six of fourteen channels met the alive criterion.

Table H.2: Per-channel verdicts, over distinct sensor updates.

Left arm		Right arm	
Channel	Verdict	Channel	Verdict
j1	intermittent, 2.2 %	j1	alive, 195°
j2	incoherent, 5 %	j2	alive, 261°
j3	incoherent, 8 %	j3	incoherent, 14 %
j4	incoherent, 6 %	j4	alive, 291°
j5	incoherent, 12 %	j5	dead
j6	incoherent, 8 %	j6	alive, 250°
j7	alive, 26°	j7	dead

Left j7 had been recorded as railed and dead: the evidence was a spread of exactly zero over forty consecutive *transmitted rows*, which is what an oversampled sensor (82 to 88 % repeated rows) produces regardless of health. Regressing true reach on the surviving channels plus the inertial sensor gives $R^2 = 0.133$ (left) and $R^2 = 0.986$ (right), naming the repair as exactly left j2 and j4.

H.3 Axis correctness

Regressing commanded pose against the master’s physical spherical position (6443 samples, $R^2 = 0.999$) found a gain matrix of $\text{diag}(+1, -1, -1)$ against the intended identity: raising the master drove the arm down. Corrected. The lateral axis remains wrong and cannot be fixed by any axis map: measured directions meant to be 90° apart are 137.7° apart, and a rotation preserves angles, so no rotation maps one triad onto the other. Gyroscope-derived azimuth halves the residual on the left arm (25.6° to 13.5°) but is not enabled: the right sensor drifts at $44^\circ/\text{min}$ stationary.

H.4 Safety: fault injection and timing

Fourteen of fourteen injected faults handled: channel dropout, board disconnect/reconnect, emergency stop mid-motion, wrong-object latching, ambiguous-target flicker, stale object, detection failure, logging failure, identifying data in a manifest, non-monotonic clock, session loss, zero detections, network dropout. Dead-man timing against a 1.00 s deadline: healthy master, 243 frames, no trip; **degraded (publishing at full rate, frozen source timestamp): trip at 0.94 to 0.99 s** – the case that could not have passed before keying the dead-man on the source timestamp; clean unplug: trip at 0.97 s. A separate emergency-stop defect: the trip handler called a blocking real-driver shutdown first, so decision at 0.97 s produced trip *logged* at 4.98 s – fixed by parking first, from local publishes only.

H.5 Task-set verification

Every coordinate of the five-task set verified at ten repeats over paths densified to 20 mm, both arms confirmed at 0.0000 rad from home before any sample: zero failures, and the result reproduces field for field on re-run.

Table H.3: Five-task verification.

Task	Coverage	Result
Positioning	3 targets per arm	3/3, 3/3
Pick and place	3 picks per arm, approach and place	3/3, 3/3
Rigid carry	3 paths, 25 densified waypoints	all verified
Compliant carry	as rigid carry	all verified
Dual pursuit	26-direction shell per arm	26/26, 26/26

H.6 Performance figures

Control-path latency; a lower bound on end-to-end latency.

The Kortex figure constrains the architecture: the cyclic control interface needs a 1 kHz loop on a dedicated link, and over this machine’s virtualised network each cyclic write costs about 10 ms – at 100 Hz the write alone consumes the cycle budget and no command reaches the arm while telemetry keeps arriving. The high-level velocity interface avoids this because it holds a commanded motion between sends.

Table H.4: Measured rates and latencies, against mock hardware.

Quantity	Value	Note
Master frame rate	100 Hz	firmware loop
Distinct sensor updates	14.7 to 17 Hz	Chapter A
VR packet rate	68.5 Hz	822 packets / 12.0 s
VR round trip	8.66 ms median, 16.5 ms p95	
Voice stop to /estop	2 ms	
Kortex high-level loop	18.4 to 18.7 Hz	two round trips/cycle

H.7 The first hardware session

Measured on a physical arm – the only section of this chapter whose numbers come from a moving Kinova arm. A single arm was driven from its own wrist camera for the first time. Three findings survive as results. **The camera-to-wrist transform is wrong by roughly 8.5° and rotates with the wrist:** the surface normal measured from four poses of a stationary table read $9.74, 15.42, 16.26, 18.14^\circ$ from vertical, a worst pairwise disagreement of 8.45° – 34 mm at 0.23 m, 59 mm at 0.40 m. **The depth sensor itself is excellent:** the support plane fits at 94.8 % inliers, 0.46 mm residual, and a 40 mm calibration cube measures $39.2 \times 38.9 \times 40.9$ mm – only the transform out of the camera frame is wrong. **The middleware was a first-class hazard:** shared memory in a state where no new process could join while running nodes continued normally, and an unset middleware variable caused silent discovery failure. Both present as an unresponsive robot.

H.8 Verification, in totals

Table H.5: What is checked automatically, as of 2026-08-31.

Measure	Count
Automated tests passing	1505
Test modules	136
Checks on the operating window	257
Faults injected and handled	14 of 14
Simulation stages, all of which ran	16
Inverse-kinematics calls, failures	13 927, 0
Known-answer checks on the vision arithmetic	21
Simulated studies of the analysis procedure	20 000
Recorded demonstration cells	17, at 8 camera angles
Vision stages instrumented on real cameras	132

H.9 What is not measured

Anything derived from a force reading (the mock returns identically zero); object detection rate at working distance (the renderer used to measure it is out of the detector’s training distribution); grasp success against a physical object; end-to-end latency including the arm and operator; how the terminal joint error varies with commanded speed (retracted in [Chapter E](#): three speeds were three replicates of one condition); speech recognition against a human speaker (no audio input device on this host; every voice figure characterises the grammar, not a recog-

niser); two arms driven simultaneously on hardware (measured at 8.1 to 12.2 Hz with send latency spiking to 519 ms, past the bridge's own watchdog).

Appendix I

Workspace characterisation and its consequences

Chapter 4 gives the headline finding and the mount fix it produced. This appendix is the chain of measurements behind it.

I.1 The two arms share no reachable workspace

Over a 63-cell grid spanning the frontal working volume, arms verified at home, no cell was reachable by both arms (Fig. 4.1 in Chapter 4). The result survives freeing the wrist yaw (raising each arm’s individual reach from 16/18 to 21/21 cells, still 0/63 shared) and additional solver retries. The right arm reaches $x \leq -0.10$ m, the left $x \geq 0.15$ m: a 0.25 m dead band, measured at $y = 0.35$ m, $z = 1.10$ m, yaw free, three repeats per point.

This holds within $x \in [-0.6, 0.6]$, $y \in [0, 0.4]$, $z \in [0.90, 1.30]$ m, with the as-built mount and the legacy home joint angles – the arms share no workspace *as mounted and as parked*, and the cause is not the mount’s rotation (proved invariant) but the arms’ asymmetric home joint angles, which are ground truth because the simulation-to-hardware bridge replays them onto the real arm.

I.1.1 The fix is buildable

A parametric sweep moving the mounts inboard, forward and down, with additional forward tilt, raises shared cells from 3 to **22 of 63** within a buildable 0.20 m mount separation (splaying the mounts outward *hurts*). Recorded and deliberately not applied at the time (changing the mount invalidates the home pose, the anchor and every clearance figure), then applied later as one deliberate pass with the whole task set re-measured:

Table I.1

Quantity	Before	After
Clearance at home	0.1610 m	0.2202 m
Nearest robot part to wearer at home	the mount (no joint moves it)	the moving chain
Task objects resting on the surface	not achievable	achieved
Pick-and-place targets off centre	0.530/0.350 m	0.290 m either side

I.2 One boundary, two independent constraints

Wearer clearance and top-down grasp feasibility, measured for different reasons, land on the same lateral boundary: clearance rises monotonically and crosses the 150 mm floor at 0.25 m; top-down grasp candidates fail entirely (0/9) at $|x| = 0.25$ m and succeed entirely (9/9) at $|x| = 0.30$ m, with collision checking disabled too – kinematic infeasibility of the wrist orientation, not obstruction.

I.3 What the pinned wrist costs at the work point

Reach under each orientation policy, walked from home, understates the cost badly because home is in free space. Walked instead from (0.4, 0.175, 1.12 m) on the working surface, with the 150 mm floor enforced throughout:

Table I.2: Mean reach under each orientation policy. The two columns disagree by a factor of seven, because reach is meaningless without the point it was measured from.

Orientation policy	from home (m)	from work point (m)
pinned to the anchor (delivered)	0.460	0.065
within a 15° cone	0.492	0.304
within a 45° cone	0.530	0.325
position only	0.547	0.363

At the point where the work happens, the delivered policy leaves the arm 65 mm of mean reach against 363 mm with the wrist free – a factor of 5.6. The mechanism is redundancy, not the wearer: no walk from the work point stops on the clearance floor, they stop because there is no solution – a fixed six-degree-of-freedom pose on a seven-degree-of-freedom arm spends the entire redundancy, and pinning the wrist removes the joint that would otherwise move the elbow clear while the hand stays put. Freeing the roll alone buys 0 mm at the work point; almost all recovery comes from the first 15° (80 % of the pinned-to-free gap), which the follower's own 15° solving cone already supplies – the reason the task set solves at all.

I.4 The centre of the working space, asked four ways

Table I.3: Four attempts to place usable work within 100 mm of the body centre line, at ten repeats.

Question	Answer
Innermost workable column	$ x = 0.325$ m (L), 0.450 m (R)
Same, wearer's arms deleted entirely	0.300/0.375 m; every column 0.125 to 0.300 m binds on the torso
Objects resting on a surface, 3360 configurations	no configuration within 100 mm of centre
Same, top-down approach	0 of 840 cells, any height 0.70-1.10 m

The constraint is the torso, not the wearer's limbs: sweeping five arm postures moves the boundary by at most 75 mm, and two postures (arms clasped behind the back, arms out to the sides) place the wearer's own limbs inside the clearance floor at home, before the robot does anything. Neither arm crosses the centre line at all – zero of ten solutions, both arms, every cross-side slot – which is why the pick-and-place task draws the *side* and lets colour follow it.

I.5 Consequences for the task set

Inter-arm handover is impossible: no control, planning or autonomy change can produce a point neither arm reaches. A bimanual task must span the dead band or exploit simultaneity. The coupled carrier's original 310 mm width placed each grip inside the dead band (10 of 25 waypoints failed once the arms were verified at home); widened to 500 mm it satisfies both boundaries, and the carrier's sag tolerance was re-derived to keep the failure threshold inside the reachable band rather than outside it, where it would measure nothing.

Appendix J

Protocol for the two-person study

This is the study [Chapter 3](#) says the system was built for. The pilot in [Chapter L](#) is a smaller, operator-only precursor; this protocol – with a separate wearer – is still the target.

J.1 The question and hypotheses

Does supplying part of the operator’s control with autonomy improve performance, and what does it cost the person carrying the arms? No prior study measures both under the same autonomy in the same trial ([Chapter 2](#)). Five hypotheses, stated with direction before any data exists: task completion time is shorter under shared autonomy (H1); operator workload is lower (H2); **wearer-reported trust is lower under shared autonomy** (H3), against the direction of H1/H2, taken from a concealed-operator study [[ZGD+26](#)] and tested here with autonomy that is genuinely a program and a wearer with a concurrent task; the wearer’s sense of agency is lower while ownership is unchanged (H4), since the two dissociate [[LSK+08](#)]; cross-arm interference is lower under shared autonomy while phase lag is not (H5), which would identify the benefit as attentional. **H3 and H4 are the hypotheses worth running the study for**; H1 and H2 exist to establish the platform can show an effect at all.

J.2 Design

Two modes (direct teleoperation, shared autonomy) crossed with five tasks, within subjects, sixteen dyads, roles fixed for the whole session. Two modes rather than four is costed, not a convenience: four modes across five tasks needs 260 to 320 min against a wearer exposure cap of 48 min for carrying 17 kg continuously; a between-subjects design across modes needs ninety-six participants for the same sensitivity, since between-subject variance here exceeds the effect sought. Mode order is counterbalanced with Williams squares, not a Latin square, because a Latin square balances position but not immediate sequence, and a task with a learning curve is sensitive to the sequence.

Grasping under direct teleoperation is not achievable: the default orientation mode pins the wrist to the anchor, the pick task needs a 169.7° rotation from it, and no master channel can command that. T2 (pick and place) therefore carries no mode contrast and is analysed as a single condition – retained because grasp success under autonomy is worth measuring, and because it is the one task where autonomy’s advantage over teleoperation on this platform is categorical rather than quantitative.

Table J.1

Task	Measures	Mode contrast
T1 positioning	Throughput vs. index of difficulty [Fit54]; single- vs. both-arm decrement	both modes
T2 pick and place	Grasp success, placement accuracy	shared autonomy only
T3 rigid carry	Coordination error as object tilt	both modes
T4 compliant carry	Coordination error as gripper separation	both modes
T5 dual pursuit	Cross-arm interference under divided attention	both modes

J.3 Measures and procedure

NASA-TLX [HS88] after every block, both participants separately (the two roles load differently, and the comparison between them is the object of the study); Borg CR10 [Bor82] exertion from the wearer between every block, since a single end-of-session rating turns a covariate into a guess; trust and embodiment [LSK⁺08] at the end of session, wearer only for embodiment. A safety observer, present throughout, holds an emergency stop and watches the wearer rather than the screen; their presence is enforced in software as well as procedurally – the safety node treats two seconds of heartbeat silence as withdrawal of consent to move. Consent is taken separately, in separate rooms.

J.4 Ethics and risk

The dominant risk is a 17 kg manipulator pair, moving under partly autonomous control, within reach of a seated person’s head and torso. Mitigations: the 150 mm margin enforced over the whole path and measured geometrically (Section C.1.1); a stop within the wearer’s own reach and a second held by the observer; twenty minutes of carried-load exposure at a time with exertion rated between blocks; order counterbalanced against fatigue confounding condition; consent restated at every break; identifying data refused in code (`write_manifest()` raises on name, email, date of birth, address or telephone number). Video recording is not part of the protocol: a video of a participant wearing this rig is not anonymous, and no measure here requires one.

Appendix K

The analysis procedure, and its behaviour on constructed data

The analysis [Chapter J](#) would be evaluated with, measured against data whose answer is chosen. No dyad data exists yet; this is a property of the procedure, not a result.

The design is within dyads: for task j , the paired difference $d_{ij} = y_{ij}^{\text{SA}} - y_{ij}^{\text{DT}}$ and its standardised mean $d_z = \bar{d}_{\cdot j} / s_{d, j}$ is the quantity the design was powered for (distinct from the standardised difference of means, which deflates every effect by $\sqrt{2}$ or more once within-dyad correlation is accounted for). Five tasks give five comparisons, corrected with Holm’s step-down procedure, which controls the family-wise error rate under any dependence structure and is uniformly more powerful than Bonferroni.

Applied to 20 000 simulated studies at the planned sixteen dyads (a true $d_z = 0.73$ effect on all five contrasts; and separately, no effect at all), through the unmodified analysis code:

Table K.1: $n = 16, m = 5, \alpha = 0.05$.

Constructed truth	one contrast, uncorrected	one contrast, Holm-corrected	some contrast in the family
$d_z = 0.73$ on all five	0.777	0.638	0.973
no effect at all	0.052	0.010	0.050

The null row is the check that could fail: a false-positive rate of 0.050 against nominal $\alpha = 0.05$ shows the correction is implemented as specified. The design’s own 80 % power figure is confirmed *uncorrected* (77.7 %) but no contrast will be reported uncorrected; after Holm, recovering the effect on a *named* task falls to 63.8 %, and restoring 80 % on a named task needs twenty dyads, not sixteen. The study is nevertheless well powered for the question it actually asks – whether shared autonomy changes performance on this platform at all, where the family-wise power is 97.3 % – so a per-task null at $n = 16$ must be reported as an interval, not an absence. Trial validity is defined in one shared module that exits non-zero when no trials survive or more than half are excluded; before that existed, a session with every trial invalidated analysed cleanly to empty means and undefined intervals, indistinguishable from a null result on inspection.

Appendix L

The pilot session, in full

[Section 4.4](#) reports the headline pilot numbers. This appendix is the full record: coverage, every session, and the supplementary measures.

Twenty-three sessions were recorded, sixteen on 2026-09-03/04 for P1–P4 and seven on 2026-09-02 for P5 (recordings/sessions/). Duration, hand-travel distance and lag come from the full-resolution position trace (20 Hz); clutch and emergency-stop counts come from each session’s event log. P5’s sessions predate the per-participant folder-naming convention (generic timestamp names, no task label) and are identified as direct or shared from the `shared_autonomy` flag each session’s own `events.jsonl` logs at start, which – unlike the folder names relied on for P1–P4 – is populated correctly for every one of P5’s sessions. Participant identity is P1–P5 throughout, including in every figure pixel, consistent with `write_manifest()` refusing to record a name ([Section 3.5](#)).

Table L.1: Which participant performed which task under which condition. Coverage is uneven because a pilot session is limited by time. P5’s task was not recorded; all seven of P5’s sessions are one undifferentiated block.

Participant	Object tracking	Target reaching	Position matching	(unlabelled)
P1	both (direct $\times 2$)	—	—	—
P2	both	both (shared $\times 2$)	—	—
P3	direct only	shared only	both	—
P4	both	—	both	—
P5	—	—	—	both (direct $\times 5$, shared $\times 2$)

L.1 Sim-to-real tracking during the pilot

Every pilot session polled the physical Kinova arms’ own joint encoders alongside the simulated joint state, which makes a direct check possible: does the real arm actually follow the simulated one it was commanded through. [Figure L.3](#) is the best-tracking session across all twenty-three (P4, object tracking, direct control, chosen by lowest sim-to-real joint RMS): the real arm’s joint 4 angle follows the simulated one to 1.06° RMS over 3859 samples, with the largest departures during the fastest motion and near-zero error while the arm holds still. This is the same terminal-offset mechanism [Section E.3](#) measures from replayed commands, seen here live during an operator session rather than in an offline replay – **it is evidence the sim-to-real bridge was live and tracking throughout the pilot**, not a claim that any pilot result was validated against a loaded arm, which requires the arm to be carrying and moving an object rather than only mirroring a joint angle.

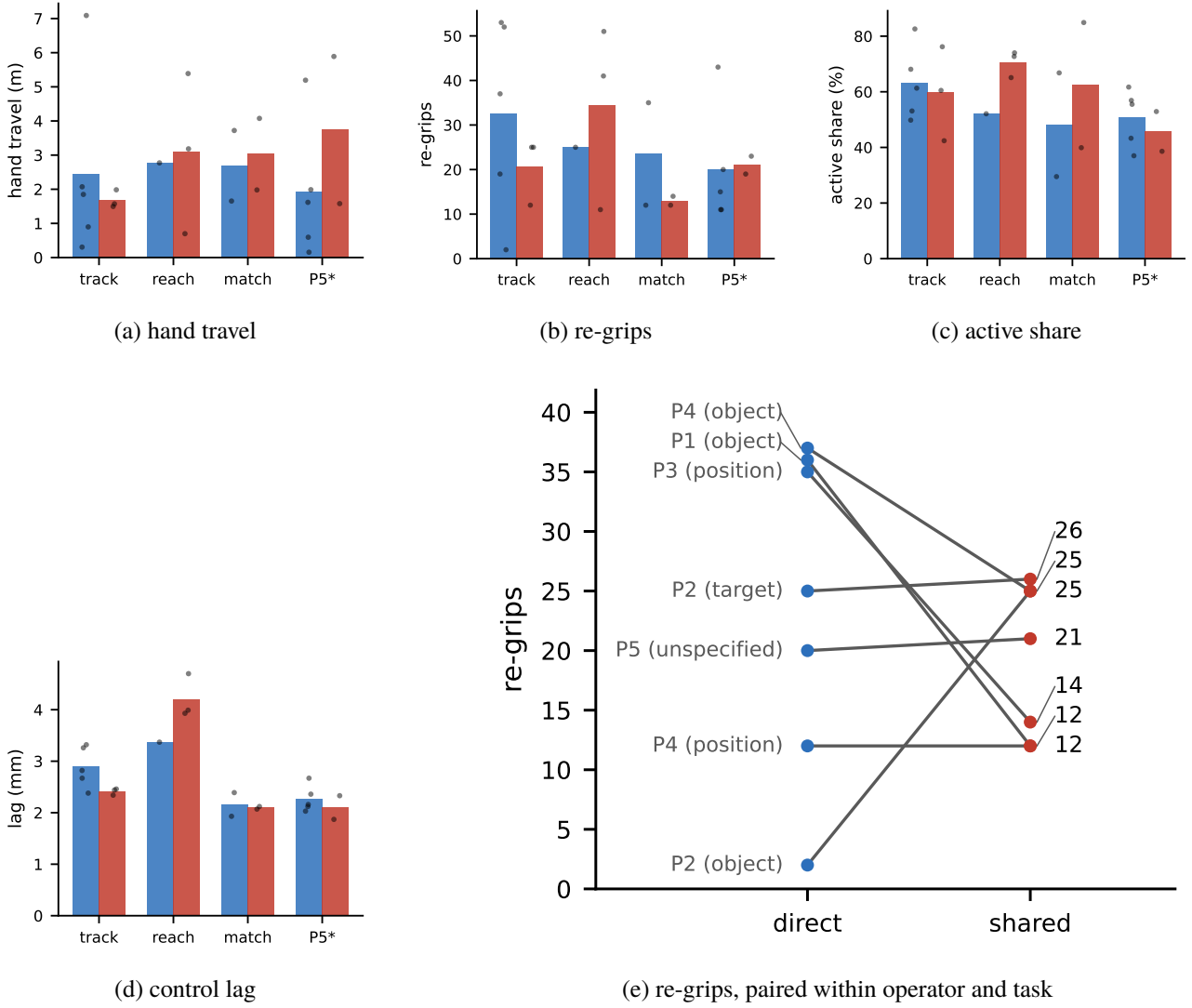


Figure L.1: The supplementary pilot measures, by task and condition. Lag is essentially unchanged between conditions; hand travel (a) is higher under shared autonomy in three of four task groups, consistent with the paired result in [Section 5.4](#). Both are drawn from the position trace rather than a stopwatch, which is why they are read as more reliable than session duration in a pilot where duration is confounded with setup time.

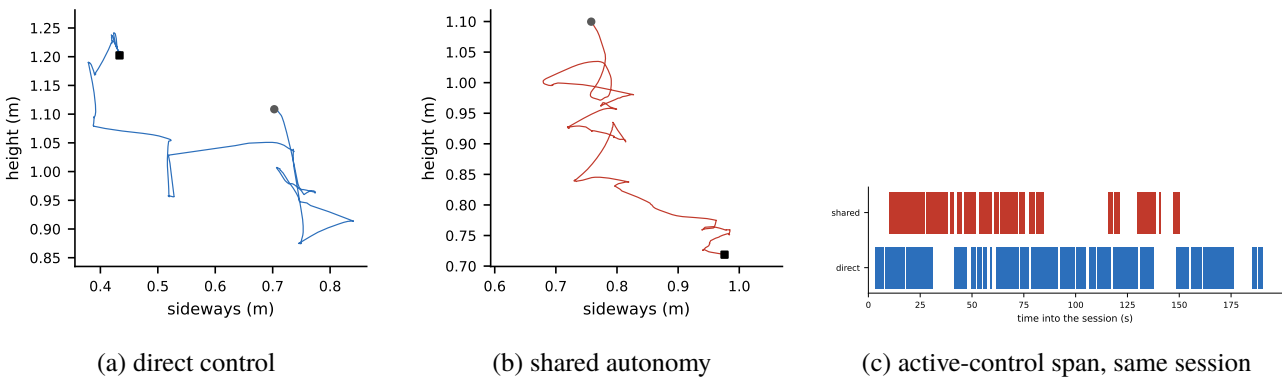


Figure L.2: The actual recorded end-effector path (`ee_left_x/y/z` in `trail.csv`) for one participant on the same task under both conditions, recorded back to back, with the corresponding span of active control alongside. Neither path is smooth; a shorter, straighter line would indicate a more direct approach to the target. Gaps in the engagement panel are the operator releasing the grip to reposition their hand before re-engaging.

Table L.2: Every recorded session. Duration includes setup and hesitation. Active share is the fraction of the session spent with the grip engaged. P1’s two direct-control sessions are a short warm-up followed by the recorded attempt, both retained as separate points.

P	Task	Cond.	Duration (s)	Travel (m)	Re-grips	Active
P1	Object tracking	Direct	147.4	0.31	19	50 %
P1	Object tracking	Direct	392.8	1.85	53	53 %
P1	Object tracking	Shared	114.9	1.50	12	61 %
P2	Object tracking	Direct	38.7	0.90	2	61 %
P2	Object tracking	Shared	169.6	1.57	25	76 %
P2	Target reaching	Direct	227.9	2.77	25	52 %
P2	Target reaching	Shared	78.6	0.70	11	65 %
P2	Target reaching	Shared	357.9	5.39	41	73 %
P3	Object tracking	Direct	617.4	7.09	52	83 %
P3	Target reaching	Shared	458.1	3.18	51	74 %
P3	Position matching	Direct	315.9	3.72	35	67 %
P3	Position matching	Shared	196.8	4.08	14	85 %
P4	Object tracking	Direct	195.6	2.07	37	68 %
P4	Object tracking	Shared	186.9	1.98	25	42 %
P4	Position matching	Direct	171.9	1.65	12	30 %
P4	Position matching	Shared	111.7	1.98	12	40 %
P5	(unlabelled)	Direct	262.5	1.99	20	37 %
P5	(unlabelled)	Shared	250.0	1.58	23	39 %
P5	(unlabelled)	Direct	359.5	5.19	43	62 %
P5	(unlabelled)	Shared	265.8	5.89	19	53 %
P5	(unlabelled)	Direct	207.8	1.62	15	43 %
P5	(unlabelled)	Direct	93.0	0.16	11	56 %
P5	(unlabelled)	Direct	99.3	0.59	11	57 %

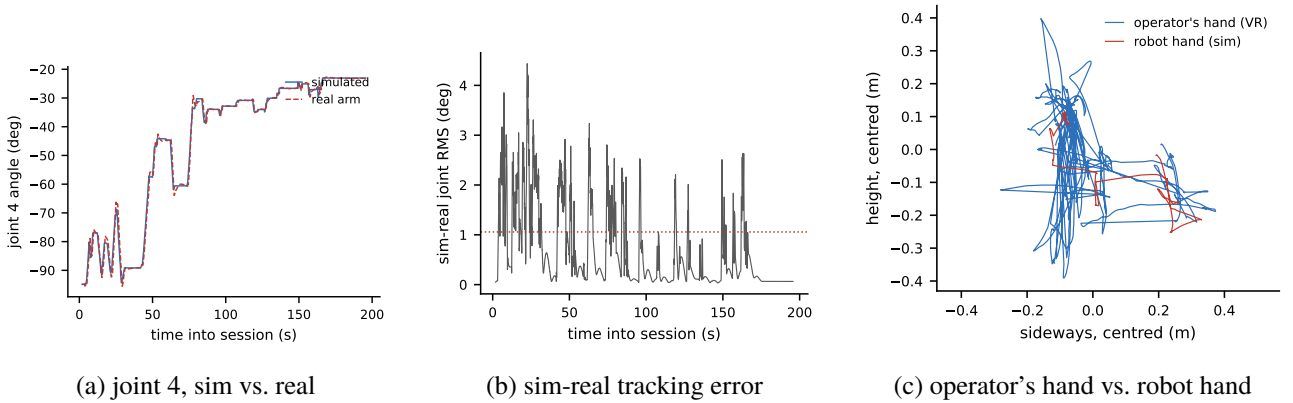


Figure L.3: The best-tracking pilot session by sim-to-real joint RMS. (a) One joint, simulated command against the real arm’s own encoder, over the session. (b) The resulting tracking error, root-mean-square over all seven joints; the dotted line is the session RMS, 1.06° . (c) The operator’s raw VR-hand path against the simulated robot hand’s achieved path, each centred on its own mean – the two live in different coordinate frames and at different scales by design, so an absolute overlay would not be meaningful; centring each on itself is the honest way to compare their shapes.

L.2 The master-arm cohort

Three further operators (M1–M3, disjoint from P1–P4) attempted the same protocol from the instrumented mannequin master rather than VR controllers, across sixteen further recording attempts on 2026-09-02 and 2026-09-04. **Seven of those sixteen are confirmed empty:** `ros2 bag info` reports zero messages, or no bag metadata at all, for every attempt at the multimeter task and both of M1’s target-reaching attempts – the same recording-pipeline failure this project’s own session index for that day logged independently (eight of twenty-six recordings clean, ten with no data at all). The nine that did record are real sessions, 72 to 304 s each, tens of thousands of messages.

Table L.3: The nine recorded master-arm sessions. Joint RMS is the sim-to-real tracking error, computed the same way as [Fig. L.3](#), over the active arm’s seven joints; see [Fig. L.4](#).

M	Task, condition	Duration (s)	Hand travel (m)	Joint RMS (deg)
M1	Object tracking, direct	154.1	21.44	21.2
M1	Object tracking, direct	145.8	14.11	15.9
M1	(unspecified), shared	209.8	22.02	16.3
M2	Object tracking, direct	110.9	10.85	2.2
M2	Object tracking, shared	71.7	7.57	1.9
M2	Target reaching, direct	303.8	20.80	1.8
M2	Target reaching, shared	83.6	4.19	0.4
M2	Target reaching, shared	206.4	11.60	3.0
M3	Target reaching, direct	254.1	11.48	1.8

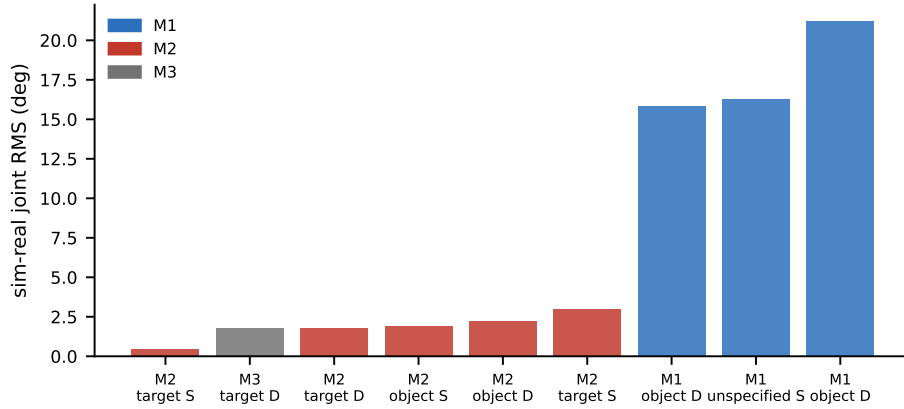


Figure L.4: Sim-to-real joint tracking error for every recorded master-arm session, sorted low to high. M2 and M3 track within 3° throughout; **every one of M1’s three sessions tracks an order of magnitude worse** (15.9 to 21.2°).

M1’s tracking error is not noise: it is consistent across all three of that operator’s sessions and separated from M2 and M3 by a factor of five to fifty. The mechanism this project’s own channel-health measurement predicts ([Chapter H](#)) is a session that leans on the incoherent left-arm channels rather than the coherent right-arm ones – all three of M1’s sessions used the *right* arm as the active hand ([Table L.3](#)), so a simple left/right split does not explain it on its own, and which specific channels M1’s sessions exercised has not been isolated. It is reported here as a finding this pilot surfaced, not a diagnosis: the master-arm modality can apparently produce sim-to-real tracking an order of magnitude worse than either VR teleoperation or another operator on the same hardware, on the same day, and the pooled VR-only comparison of [Section 4.4](#) would have missed it entirely.

Limits. Five participants with uneven task coverage is a pilot-scale sample: no significance test is reported, and none of the differences above should be read as a proven effect. The pooled figures mix three tasks together; the paired panels are the part of this pilot closest to the paired design of [Chapter J](#). Session duration runs from the start of the recording to the end, including any explanation or hesitation, so it is a session-length measure rather than a pure motor-execution time. Every session's emergency stop was engaged once, near the end, to stop the recording – a procedural marker rather than a safety incident – and is excluded throughout. This pilot has no separate wearer: every measure here is what the operator experienced, and the wearer-side hypotheses of [Chapter J](#) (H3, H4) remain untested.

Appendix M

Background, in full

Chapter 2 states the gap this project addresses. This is the full review behind it, organised by theme, with each citation's scope stated precisely rather than implied.

M.1 Supernumerary robotic limbs

The field's founding demonstrations are Parietti and Asada's bracing limbs for aircraft fuselage assembly, later extended to whole-body support [PA16], which anchor against the environment so the reaction load leaves the wearer's body rather than passing through it (Fig. M.1). In the arrangement studied here the arms are free in space, so the chain closes through the wearer instead – and every reaction load is carried by a person who is not the person commanding the motion. This difference, not the control law, is what makes wearer clearance the binding constraint of Chapter I.

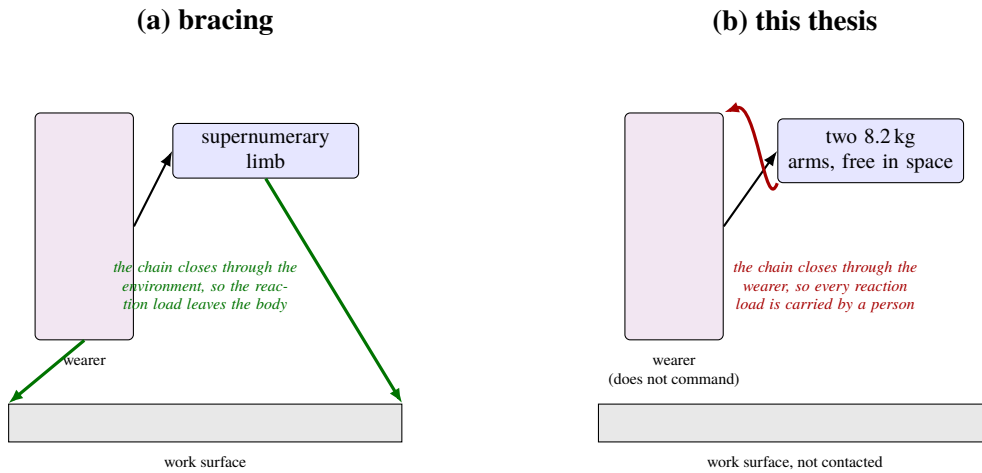


Figure M.1: Bracing (the field's founding configuration [PA14]) against free-space mounting (this platform).

Reviews since map the design space along consistent axes – mounting location trades reach against the moment arm imposed on the wearer, actuation trades bandwidth against worn mass [YHC⁺21, P⁺21, TL21, Z⁺23] – and a recent quantitative survey adds that the field reports embodiment far more often than task performance, and rarely both [Z⁺26]. Worn mass is the constraint reported least: each Gen3 masses 8.2 kg, the pair with harness exceeds 17 kg, and simulated wearer loads suggest the biomechanical cost is not negligible [MK24].

M.2 The operator and the wearer as different people

Fusion mounts a robotic head and two arms on a worn frame, driven by a remote operator [SSM⁺18]. Two qualifications travel with that citation: its three modes (Direct, Induced, Enforced) are levels of *physical coupling* between the robot arms and the wearer’s own, not levels of machine autonomy – in all three the robot decides nothing – and the publication is a two-page demonstration with no user study and no performance numbers. It establishes that the architecture existed; it cannot support a claim about how well it worked.

Work on spatial guidelines for near-body interaction ran a concealed operator, varied apparent autonomy, and measured the wearer’s physiological arousal and self-reported trust [ZGD⁺26]: higher apparent autonomy made both worse. Their autonomy was a hidden human and their wearer had no task of their own, so the result does not settle what happens when autonomy is real and the wearer is occupied – it establishes only that the wearer’s experience is not a free variable. Disturbance rejection for the two-person case is partly solved: Zhang et al. compensate human-induced base motion and report end-effector errors of (1.37 ± 0.58) mm [Z⁺24], with the human shoulder as a floating base – covering the case where wearer and operator are the same person (the base motion is partly predictable to the controller because the person commanding it also caused it), not the case studied here.

M.3 Master interfaces for dissimilar kinematics

Guggenheim and Asada argue for miniature physical masters on the grounds that geometric similarity supplies feedback inherently, without a separate haptic display [GA21]. The argument covers similarity of *form*; this system’s master-to-slave reach ratio is roughly one to three, which is where it departs. Joint-space correspondence preserves posture but needs every master joint measured accurately; task-space correspondence decouples the chains at the cost of the slave’s redundant degree of freedom (Chapter B). Pinardi et al. review supplementary feedback in movement augmentation and find that feedback improving control does not reliably improve the sense of ownership [PLF⁺23] – a caution against treating either as a proxy for the other.

M.4 Shared autonomy

Dragan and Srinivasa formalise arbitration as policy blending: a predicted user goal and the user’s actual input combined with a weight set by prediction confidence [DS13], explicit that blending aggressively on a wrong prediction is worse than not blending. Javdani et al. recast the problem as a POMDP over the user’s goal, approximated by hindsight optimisation [JSB15]: their user study found assistance that does not wait for high confidence lets operators finish faster with less input. Neither line of work was developed for a robot mounted on a second person – there, the cost of assisting toward the wrong goal is a manipulator moving toward someone who did not command it, not a slower trial.

M.5 Measuring embodiment, workload and trust

The rubber hand illusion showed ownership can be induced over an object that is plainly not one’s hand [BC98]; Longo et al. decompose the resulting experience into ownership, agency and location, which can move independently [LSK⁺08] – an operator may feel authorship of a motion without feeling the arm is part of them, and a wearer may feel neither while bearing the consequences. Workload is conventionally the NASA Task Load

Index [[HS88](#)]; for a two-person system it must be administered to both, since the two will not load the same way. Physical exertion is better tracked with a Borg category-ratio scale [[Bor82](#)]. ANSI/ISA-101.01's central principle for operator displays [[Int15](#)] – colour is a scarce alarm channel, so a normal state should be largely uncoloured – transfers directly to a robot console.

M.6 The gap, restated precisely

Fusion established the architecture and measured nothing. Shared autonomy raises operator performance where the robot is attached to nobody. Raising apparent autonomy lowers a wearer's trust where the autonomy is a concealed human and the wearer has no task. What has not been done is to measure both ends in one system, under the same autonomy, in the same trial – the engineering contribution of this thesis is a system in which that question could be asked; [Chapter 5](#) sets out what still stands between the system as built and the full session that would ask it.

Appendix N

Discussion, extended

Chapter 5 is the argument. This appendix adds the detail that did not fit there.

N.1 What the degradation architecture bought, and what it did not

The degradation architecture kept the system running and its failures visible: with eight of fourteen channels failing ([Section H.2](#)), the right arm retains full position sensing and the left retains direction, and the incoherent verdict exists because a failing potentiometer does not go quiet – range alone calls a wide-spanning nonsense signal healthy. What it did not buy is capability: freezing the left arm’s bend joints collapses its commanded set from a shell to a surface, and no health scheme restores a measurement never taken. The reduced-degree-of-freedom premise succeeded exactly where a single sensor observes the quantity directly (elevation from gravity, reach from a forward-kinematic magnitude) and failed where it does not: the spherical model assumes azimuth and elevation are independent, and they are not, because the first joint’s axis does not tilt as the arm bends – a design rule for a successor, not a defect to patch with better calibration.

N.2 The clearance model itself is not yet trustworthy

The most uncomfortable single finding of the hardware programme belongs here rather than in the results. Asked about a pose in which the arm was *physically against the mannequin*, one clearance model – the one that samples link *origins* rather than link geometry, and which runs on the physical arm during homing – returned 367.0 mm, the same value it returns for a visibly clear pose. Every clearance figure in this report computed through the guard’s capsule model stands (the majority); but the standing claim that wearer margin is enforced *at all times* was, at the time of writing, **not enforced in code on hardware**. This is the one item in [Chapter 6](#)’s further-work list that is a safety defect rather than an incompleteness, and it is placed above the mount change and above the study for that reason.

N.3 Three things the real cameras changed

Beyond the headline numbers of [Chapter F](#): the surface-fitting stage transferring to hardware within a factor of three of its simulated figure is the one component of the perception chain now known to work outside simulation, and the one the pick path leans on hardest. The scene camera is a *different* instrument from the wrist cameras, not merely a distant one – an eighth of the depth returns, a twentieth of the plane support – and no autonomous behaviour should expect a support surface from it. And the learned detector contributed nothing the classical

colour-and-geometry path did not; no result in this report rests on a learned detector, worth noting given that a language-model-free, learned-detector-optional design was argued for on cost grounds rather than accuracy grounds.

N.4 Comparison against the planning report, in full

Divergences fall into three groups. *Superseded by measurement*: the haptic objective ([Chapter 5](#)). *Deferred and still open*: coordinated pick-and-place on the real platform – the task set is specified and verified geometrically and the runner exists, but the arms were not available for the period in which it would have run. *Added*: the safety architecture, the degradation architecture, the workspace characterisation and the pilot were not in the plan; the first two follow directly from discovering that sensor faults dominate, the third from asking whether the task set was reachable before running it.

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